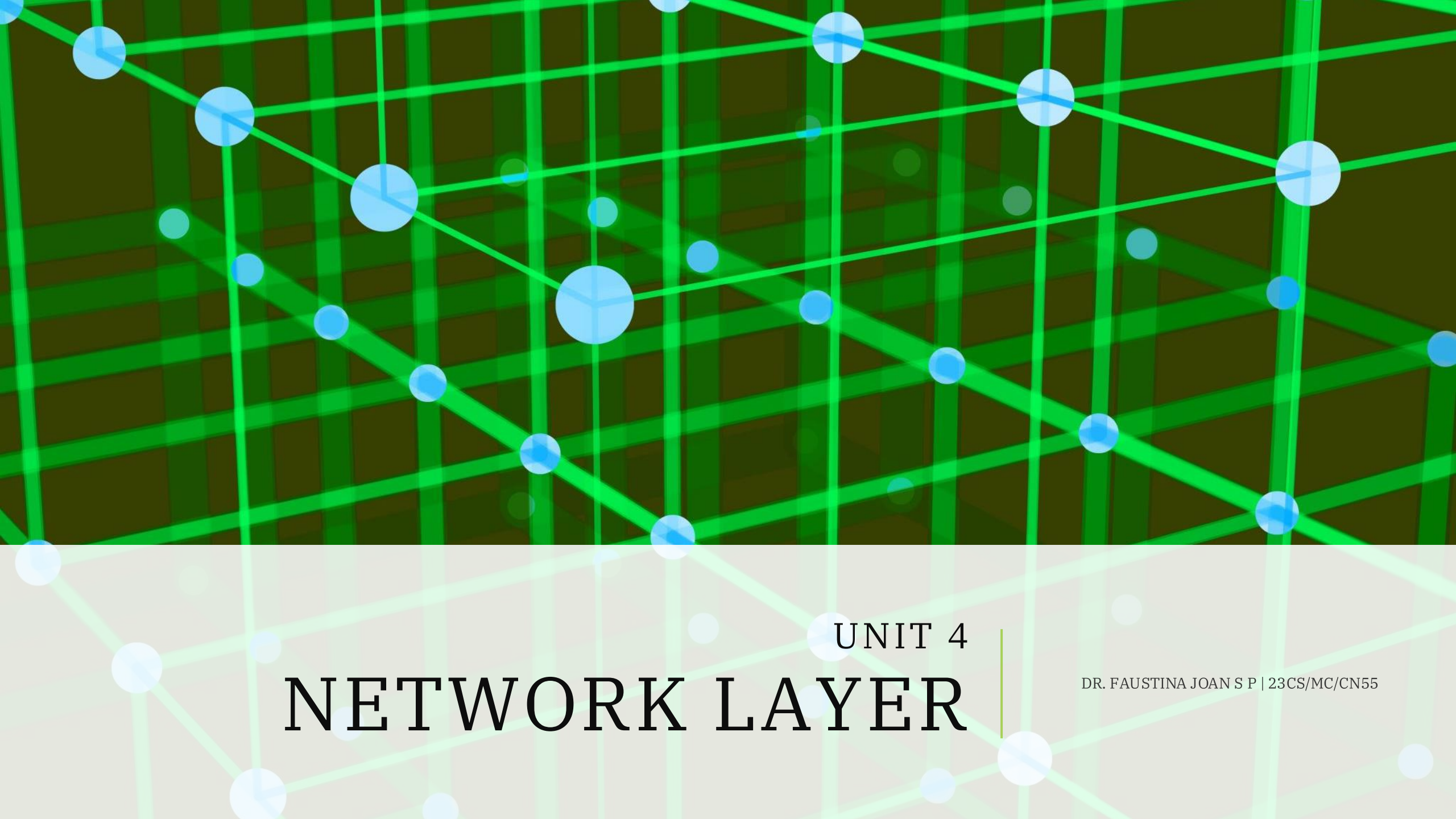


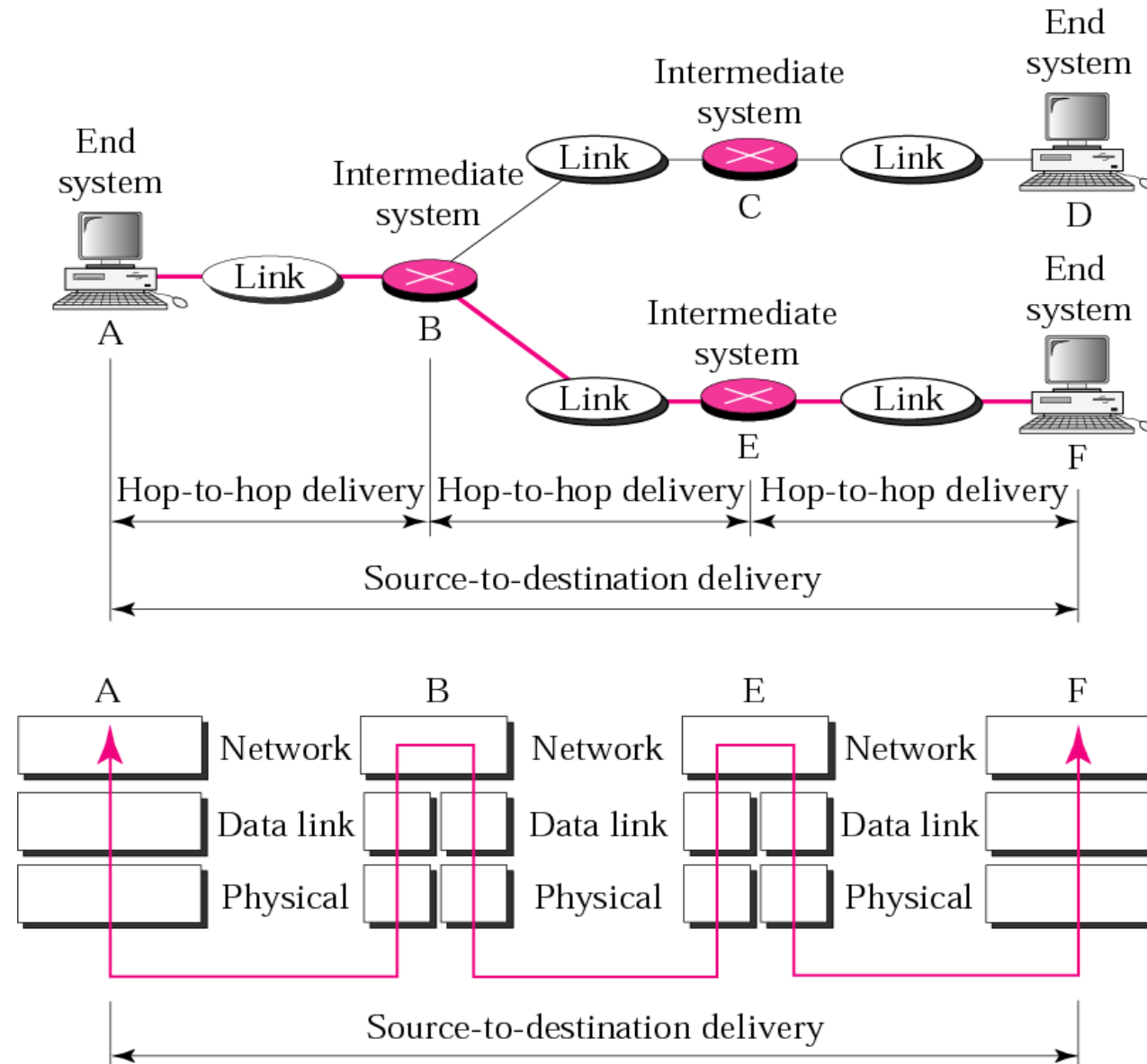


UNIT 4

NETWORK LAYER

DR. FAUSTINA JOAN S P | 23CS/MC/CN55

NETWORK LAYER



NETWORK LAYER

- The packet transmitted by the sending may pass through several LANs or WANs before reaching the destination computer.
- For this, we need a global addressing scheme, that uses logical address in the network layer of TCP/IP suite.
- The term – **IP address**



IP ADDRESS

- An **IP address (Internet Protocol address)** is a unique numerical label assigned to every device connected to a computer network that uses the **Internet Protocol (IP)** for communication.
- Identifies a device on the network (like your phone, laptop, or server).
- Provides the device's position in the network so data can be routed correctly.



TYPES OF IP ADDRESSES

- **Public IP** → Used on the internet, unique worldwide.
- **Private IP** → Used inside local networks (e.g., 192.168.x.x, 10.x.x.x).
- **Static IP** → Permanently assigned to a device (like for a server).
- **Dynamic IP** → Assigned temporarily by a network (common for home users).



IPV4 ADDRESSES

- A **32-bit address** that **uniquely and universally** defines the connection of a host or a router to the Internet.
- The IP address is the address of the connection, not the host or the router.
- Because if the device is moved to another network, the IP address may be changed.



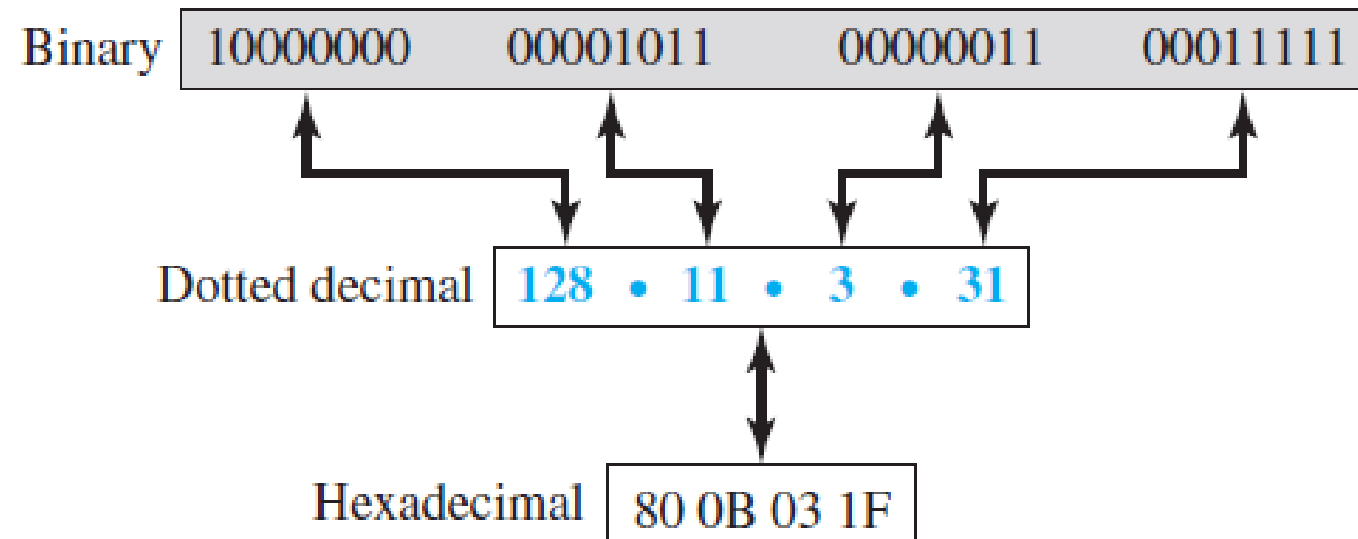
IPV4 ADDRESS SPACE

- The total number of addresses used by the protocol.
- If a protocol uses b bits to define an address, the address space is 2^b .
- IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than four billion).



NOTATION

- Binary notation (base 2)
- Dotted-decimal notation (base 256)
- Hexadecimal notation (base 16)



TWO PARTS IN AN IP ADDRESS

- Network Part / Prefix
- Host Part / Suffix



Internet as a Big City

Network Part: Prefix

Host Part: Suffix

Street Address



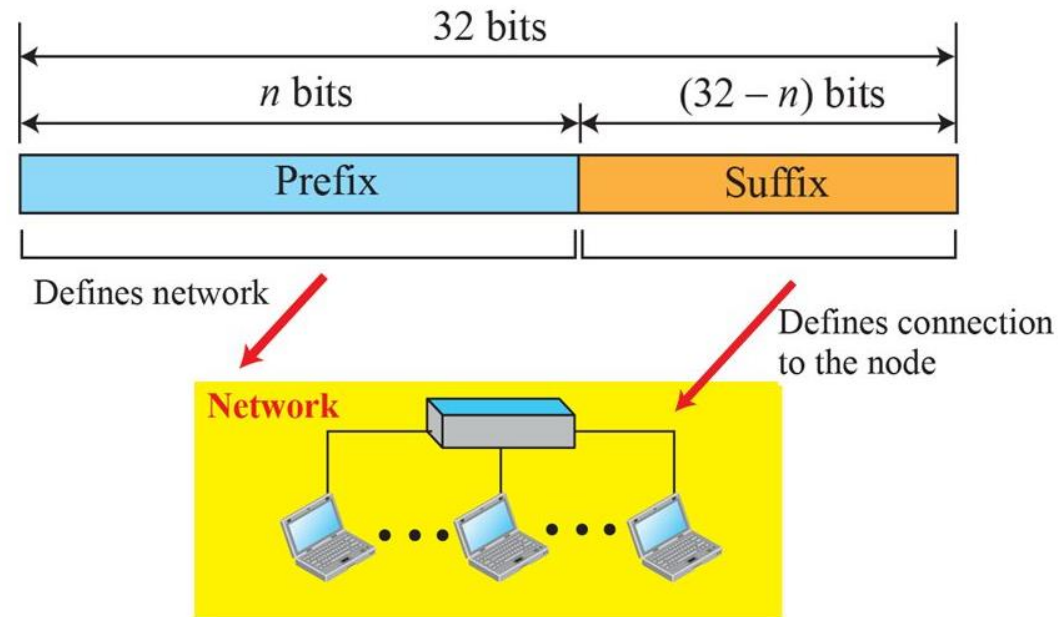
House Number



House Address

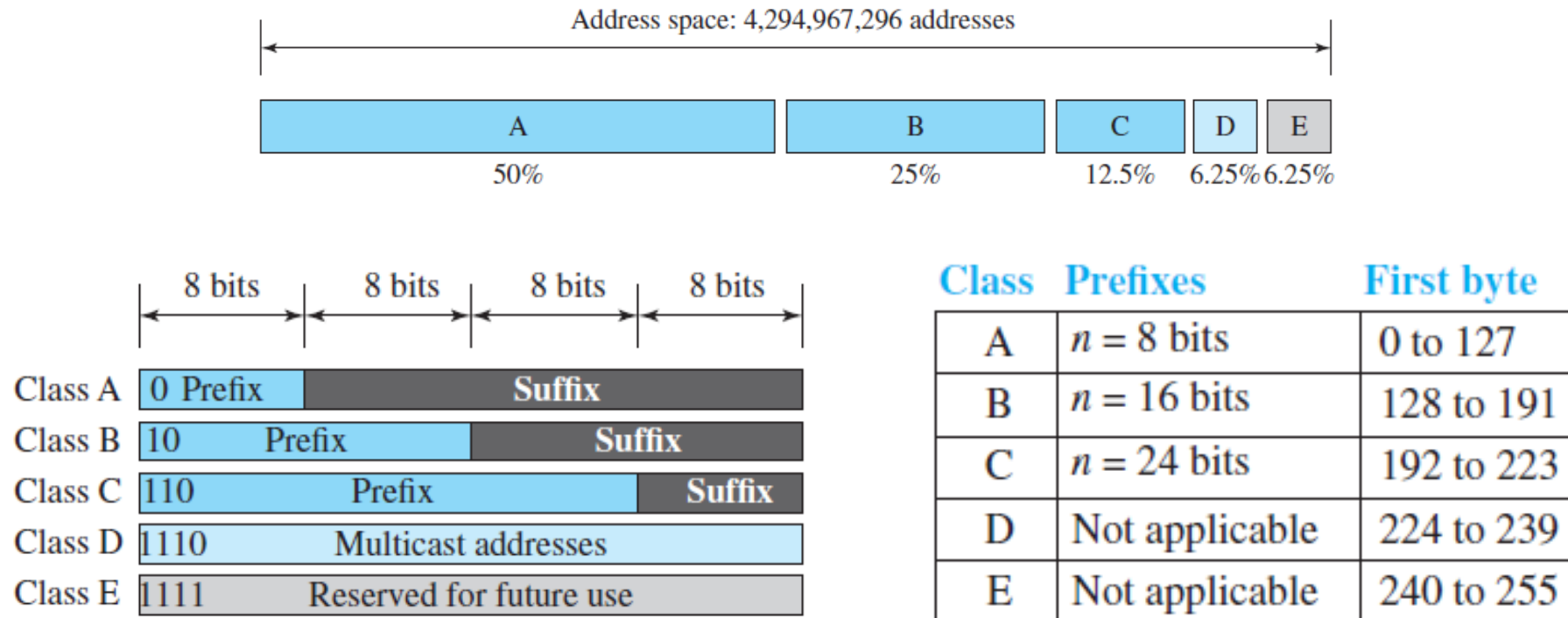
Complete IP Address

HIERARCHY IN ADDRESSING



- The network identifier in the IPv4 was first designed as a fixed-length prefix – classful addressing.
- The new scheme, which is referred to as classless addressing, uses a variable-length network prefix.

CLASSFUL ADDRESSING



- In class A, $2^7 = 128$ networks in the world can have this
- In class B, $2^{14} = 16,384$ networks
- In class C, $2^{21} = 2,097,152$ networks

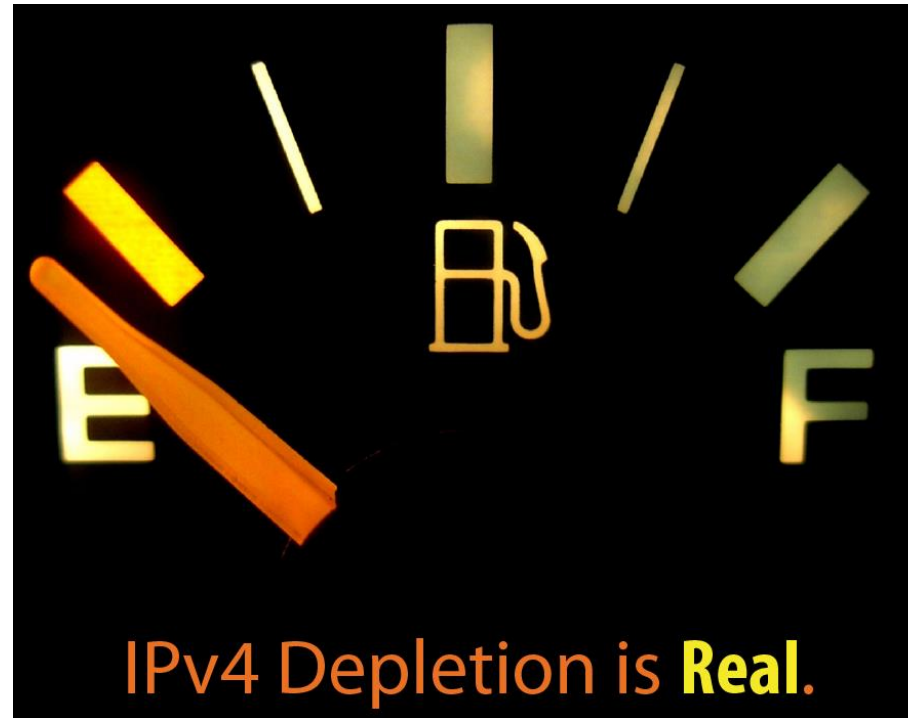


ADVANTAGE OF CLASSFUL ADDRESSING

- It was easy to find the class of the address and, since the prefix length for each class is fixed, we can find the prefix length immediately.

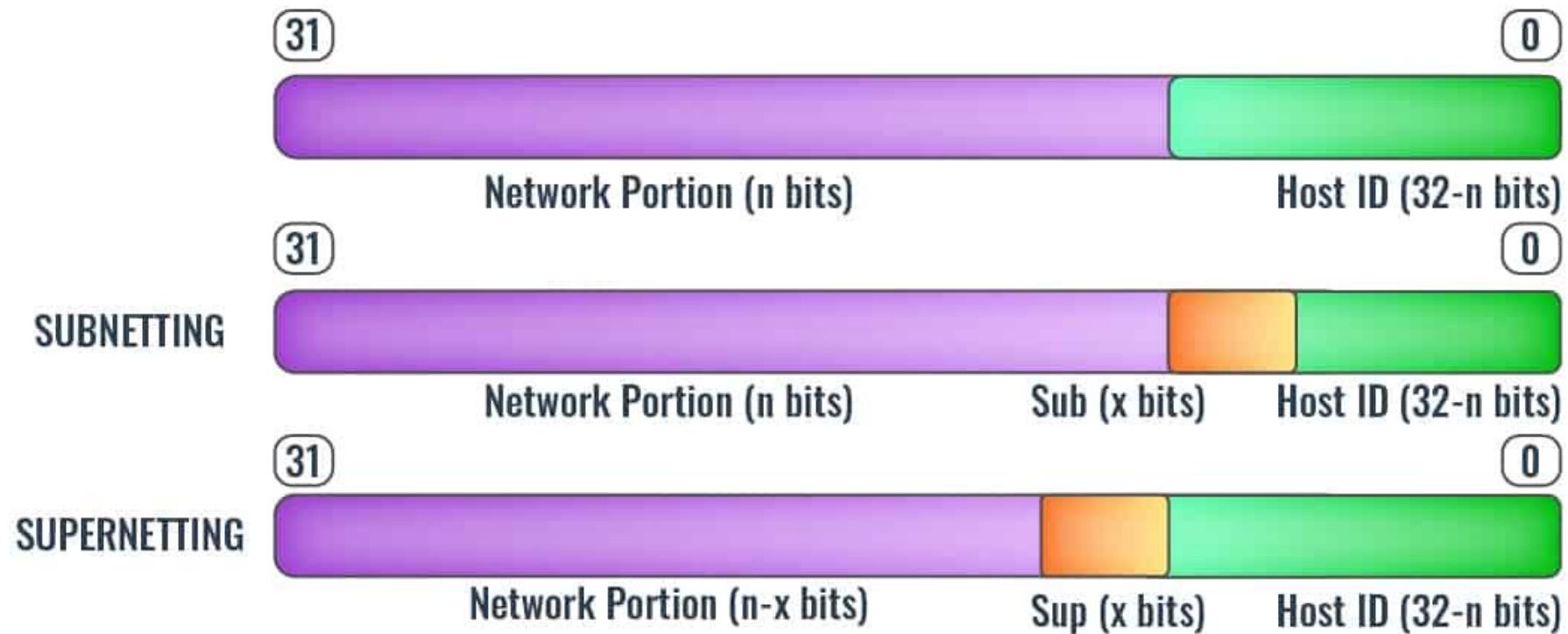


ADDRESS DEPLETION

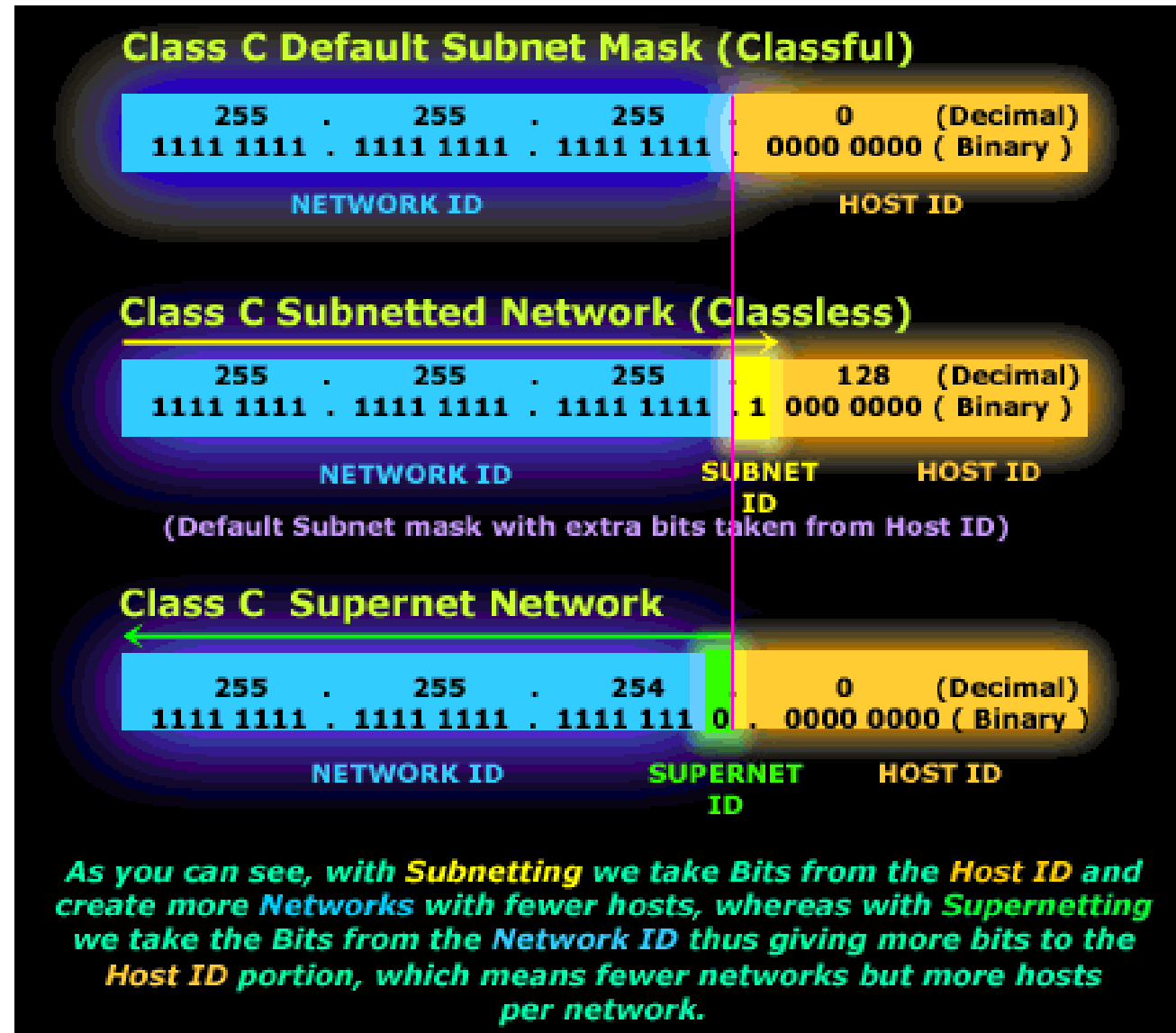


- Example: Class A can be assigned to only 128 organizations in the world, but each organization needs to have a single network with 1,67,77,216 nodes.

SUBNETTING AND SUPERNETTING



SUBNETTING AND SUPERNETTING



CLASSLESS ADDRESSING

- A short-term solution to still use IPv4 addresses, called *classless addressing*.
- The class privilege was removed from the distribution to compensate for the address depletion.
- The whole address space is divided into variable length blocks.



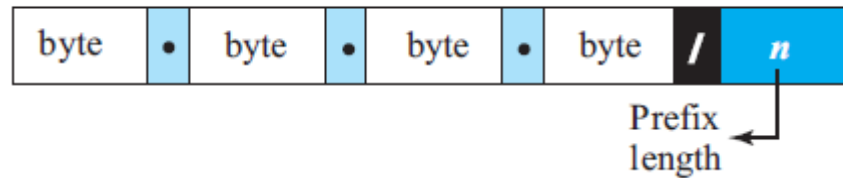
CLASSLESS ADDRESSING

- The prefix in an address defines the block (network); the suffix defines the node (device).
- The prefix length in classless addressing is variable.
- A small prefix means a larger network; a large prefix means a smaller network.



PREFIX LENGTH: SLASH NOTATION

- The prefix length, n , is added to the address, separated by a slash.



Examples:

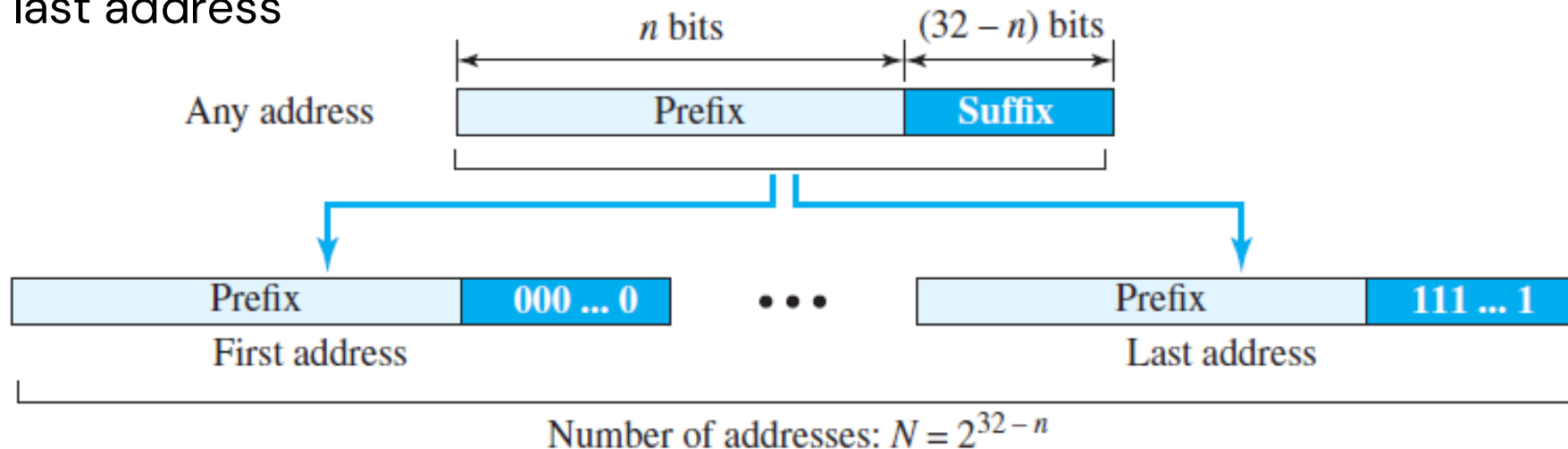
12.24.76.8/8
23.14.67.92/12
220.8.24.255/25

- Informally referred to as *slash notation* and formally as **classless interdomain routing** or **CIDR**



EXTRACTING INFORMATION FROM AN ADDRESS

- The three pieces of information tell us exactly which group of addresses the block contains.
- Given any address in the block, we can know the
 - the number of addresses
 - the first address in the block
 - the last address



EXTRACTING INFORMATION – ADDRESS MASK

- A 32-bit number in which the n leftmost bits are set to 1s and the rest of the bits $(32 - n)$ are set to 0s.
- Number of addresses in the block $N = \text{NOT}(\text{mask}) + 1$.
- First address = (Any address) **AND** (mask)
- Last address = (Any address) **OR** [(**NOT** (mask))]



ADDRESS MASK - EXAMPLE

One of the addresses in a block is 167.199.170.82/27. Find the number of addresses in the network, the first address, and the last address.

Solution

11111111.11111111.11111111.11100000

The value of n is 27. The network mask has twenty-seven 1s and five 0s. It is 255.255.255.240.

a. The number of addresses in the network is $2^{32-n} = 32$. $2^5 = 32$

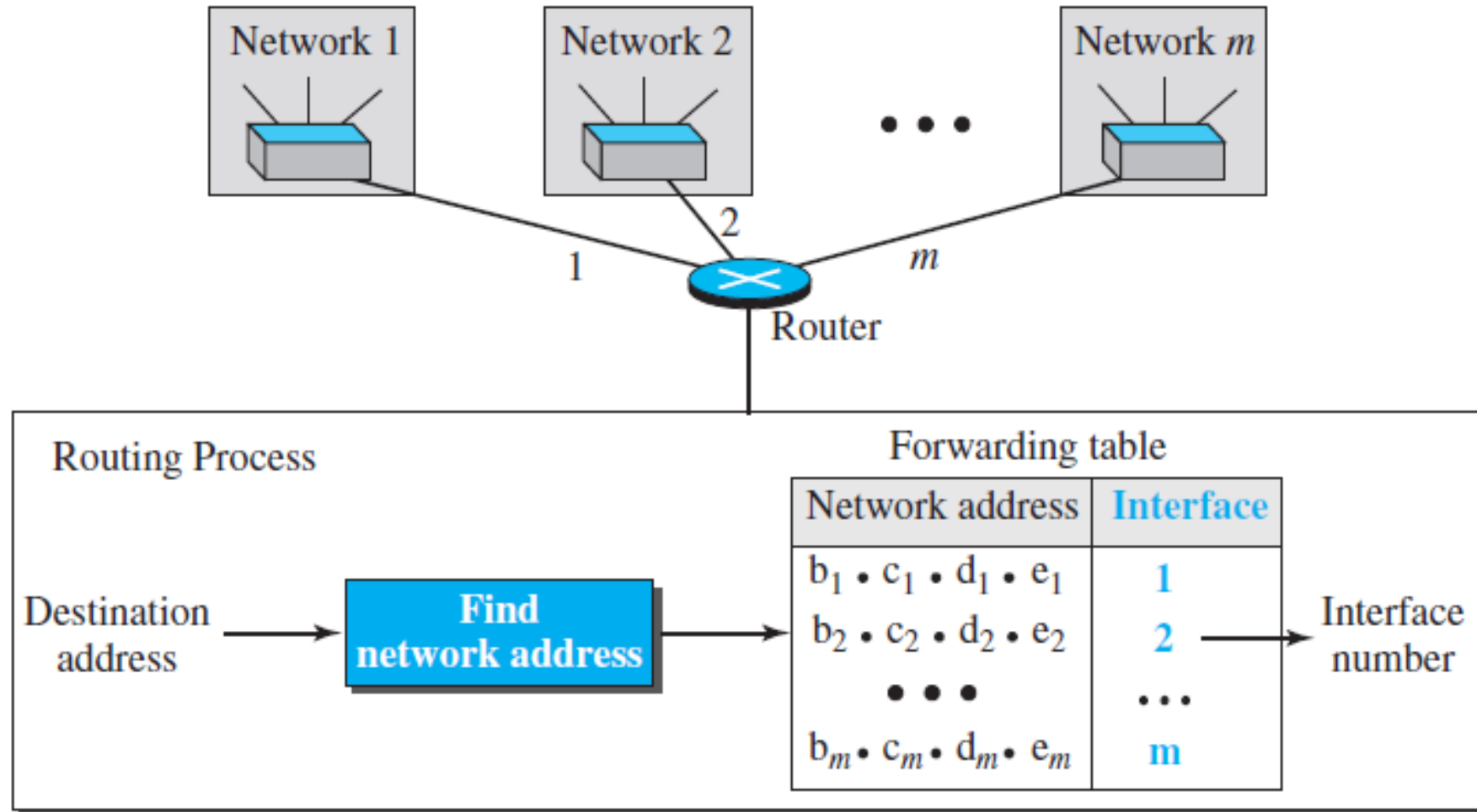
Number of addresses in the block: $N = \text{NOT}(\text{mask}) + 1 = 0.0.0.31 + 1 = 32$ addresses

Address in binary:	10100111 11000111 10101010 01010010
Network mask:	11111111 11111111 11111111 11100000
First address:	10100111 11000111 10101010 01000000

b. We use the AND operation to find the first address (network address). The first address is 167.199.170.64/27.



NETWORK ADDRESS The first address in a network / subnet



BLOCK ALLOCATION

- Done by a global authority – Internet Corporation for Assigned Names and Numbers (ICANN).
- It manages the global IP address space and allocates large blocks to Regional Internet Registries (RIRs).
 - It assigns a large block of addresses to an ISP.



BLOCK ALLOCATION - RESTRICTIONS

- The number of requested addresses, N , needs to be a power of 2.

The reason is that $N = 2^{32-n}$

- If N is not a power of 2, we cannot have an integer value for n .
- The requested block needs to be allocated where there is an adequate number of contiguous addresses available.



FIRST ADDRESS - RESTRICTIONS

- The first address needs to be divisible by the number of addresses in the block.
- The first address needs to be the prefix followed by $(32 - n)$ number of 0s.
 - Even if you find 8 consecutive addresses, you cannot necessarily choose any starting address you want.
 - For 8 addresses, the starting address must be divisible by 8.
 - So valid starting values in the last octet include: 0, 8, 16, 24, 32, 40, 48, ...
 - 192.168.1.16/29 is valid first address and 192.168.1.16 – 192.168.1.23 are the 8 addresses
 - 192.168.1.16/29 = 11000000.10101000.00000001.00010000 ✓
 - 192.168.1.18/29 = 11000000.10101000.00000001.00010010 ✗



SUBNETTING

- Assume

- The total number of addresses granted to the organization is N
- The prefix length is n
- The assigned number of addresses to each subnetwork is N_{sub} The prefix length for each subnetwork is n_{sub}

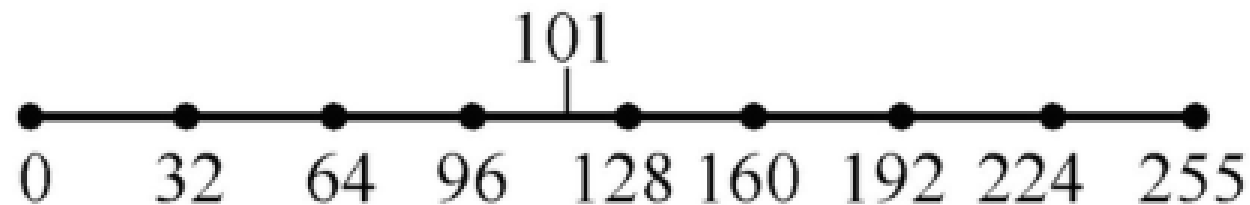
- Steps:

- The number of addresses ($N_{\text{sub}} = 2^{32-n_{\text{sub}}}$) in each subnetwork should be a power of 2.
- The prefix length for each subnetwork should be found using the following formula:
$$n_{\text{sub}} = 32 - \log_2 N_{\text{sub}} \quad (\text{Note: } 2^{32-n} \geq N_{\text{sub}})$$
- The starting address in each subnetwork should be divisible by the number of addresses in that subnetwork



SUBNETTING EXAMPLE

192.168.0.101/27



Host address: 192.168.0.101

Host address: 11000000 10101000 00000000 011|00101

Subnet mask: 11111111 11111111 11111111 111|00000

Subnet address: 11000000 10101000 00000000 011|00000

Subnet address: 192.168.0.96 All host bits are set to 0



ADDRESS AGGREGATION

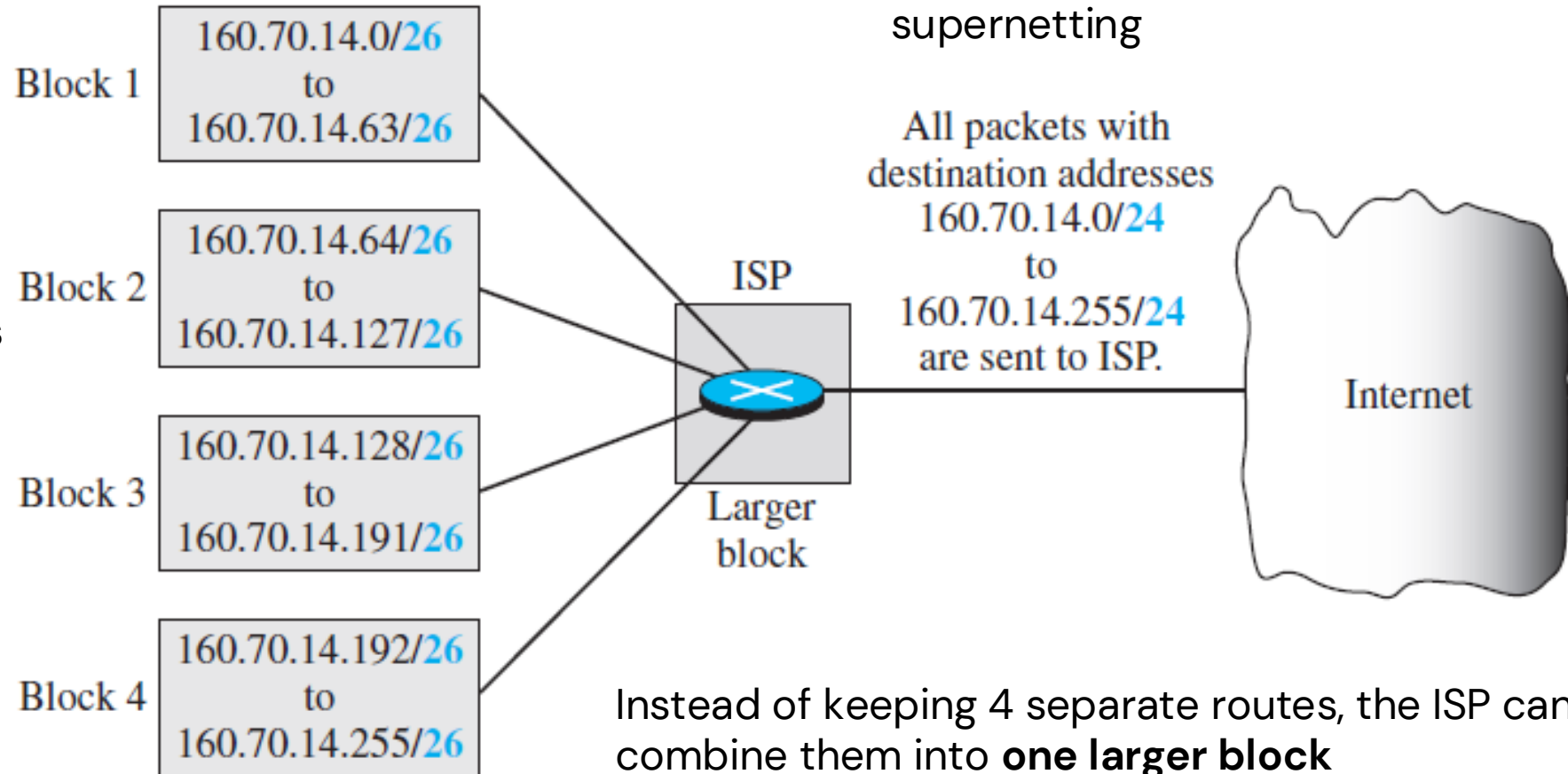
$2^6 = 64$ addresses per block

$2^8 = 256$

prefix length $32 - 8 = 24$ bits

supernetting

Total: 256 addresses
($256 = 2^8$ hosts)
suffix length – 8 bits



Instead of keeping 4 separate routes, the ISP can combine them into **one larger block**
It is mainly done to reduce the size of routing tables



SPECIAL ADDRESSES

○ ***This-host Address***

- The only address in the block **0.0.0.0/32**
- Used whenever a host needs to send an IP datagram but it does not know its own address to use as the source address.

○ ***Limited-broadcast Address***

- The only address in the block **255.255.255.255/32**
- It is used whenever a router or a host needs to send a datagram to all devices in a network.



SPECIAL ADDRESSES

○ ***Loopback Address***

- The block **127.0.0.0/8**
- A packet with one of the addresses in this block as the destination address never leaves the host

○ ***Private Addresses***

- Four blocks are assigned as private addresses: **10.0.0.0/8**, **172.16.0.0/12**, **192.168.0.0/16** and **169.254.0.0/16**.

○ ***Multicast Addresses***

- The block **224.0.0.0/4** is reserved for multicast addresses.



DYNAMIC HOST CONFIGURATION PROTOCOL (DHCP)

- Address assignment in an organization can be done automatically using DHCP.
- An application-layer program, using the client-server paradigm, that actually helps TCP/IP at the network layer.
- Often called plug-and-play protocol.
- A network manager can configure DHCP to assign permanent IP addresses to the host and routers.
- DHCP can also be configured to provide temporary, on demand, IP addresses to hosts.



DHCP - DORA PROCESS

- **Discover** – Client broadcasts a request for an IP address.
 - **Offer** – DHCP server replies with an available IP and configuration.
 - **Request** – Client requests to use that offered IP.
 - **Acknowledgement (ACK)** – Server confirms and leases the IP to the client.
-
- When a device joins, DHCP gives it a valid IP and tells it how to connect to the network, without any manual setup.



DHCP MESSAGE FORMAT

0	8	16	24	31
Opcode	Htype	HLen	HCount	
Transaction ID				
Time elapsed		Flags		
Client IP address				
Your IP address				
Server IP address				
Gateway IP address				
Client hardware address				
Server name				
Boot file name				
Options				

Fields:

Opcode: Operation code, request (1) or reply (2)

Htype: Hardware type (Ethernet, ...) 1 for Ethernet

HLen: Length of hardware address 6 for MAC address

HCount: Maximum number of hops the packet can travel

Transaction ID: An integer set by the client and repeated by the server

Time elapsed: The number of seconds since the client started to boot

Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used

Client IP address: Set to 0 if the client does not know it

Your IP address: The client IP address sent by the server

Server IP address: A broadcast IP address if client does not know it

Gateway IP address: The address of default router

Server name: A 64-byte domain name of the server

Boot file name: A 128-byte file name holding extra information

Options: A 64-byte field with dual purpose described in text



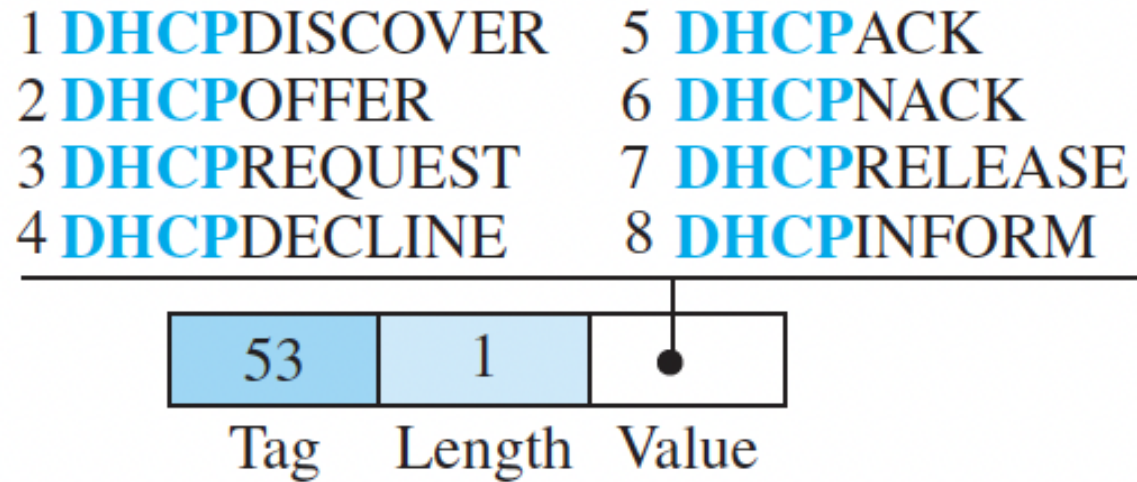
MAGIC COOKIE

- The server uses a number, called a magic cookie, in the format of an IP address with the value of 99.130.83.99.
- When the client finishes reading the message, it looks for this magic cookie after BOOTP message (upto 236 bytes till boot file name)
 - BOOTP was an older protocol used to help a computer get its network configuration when it starts up.
 - DHCP was later developed from BOOTP and uses much of the same basic message structure.
- If present, the next 60 bytes are options.



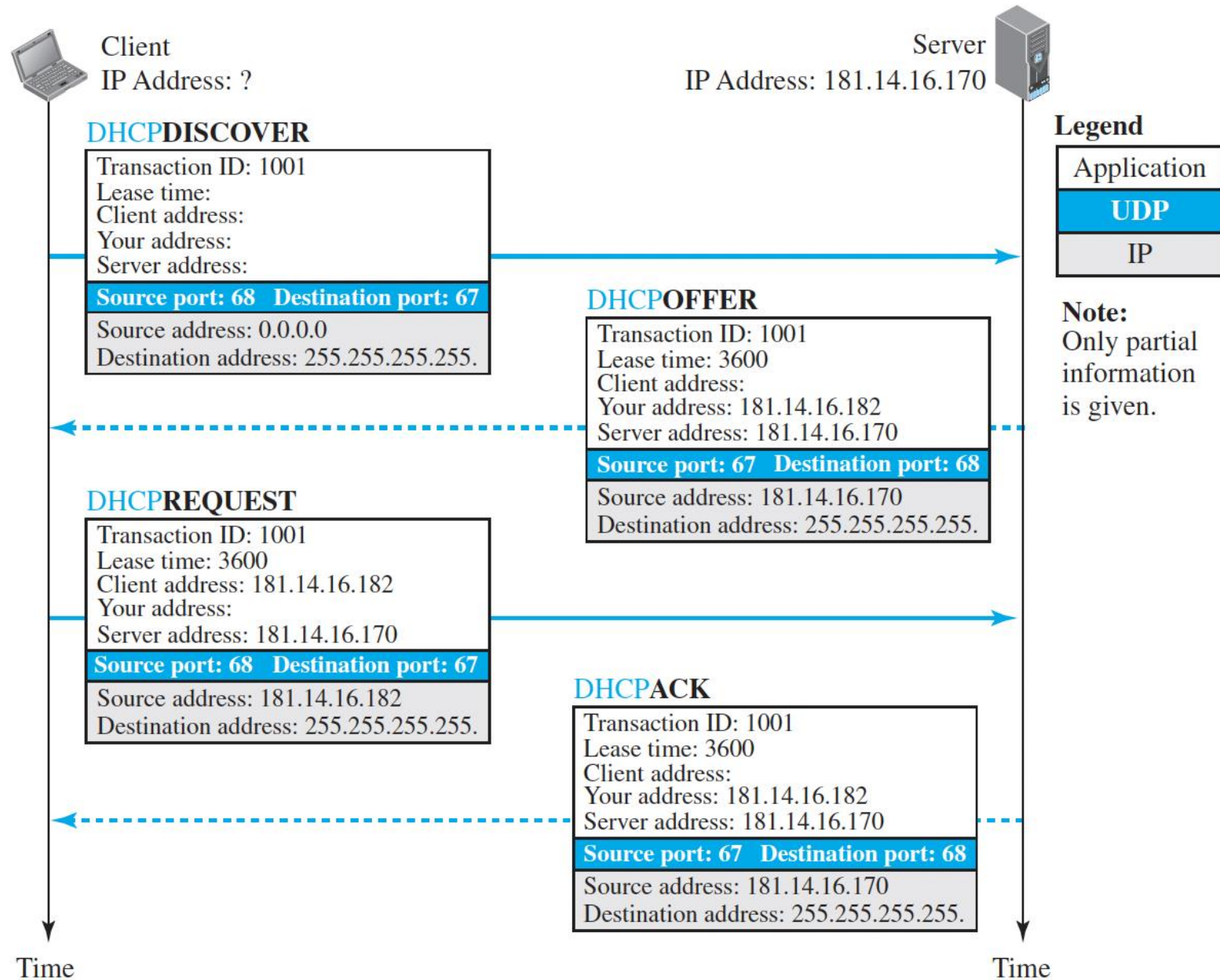
OPTIONS

- An option is composed of three fields: a 1-byte tag field, a 1-byte length field, and a variable-length value field.

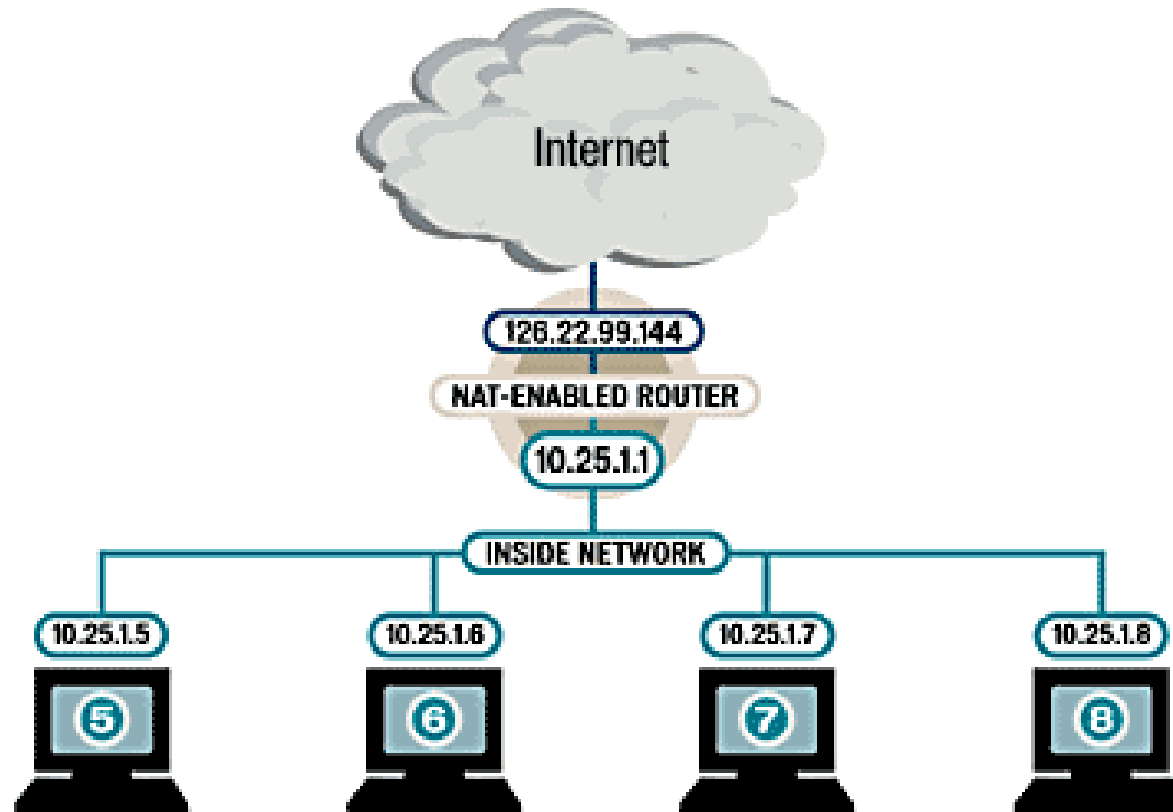


53 → DHCP Message Type



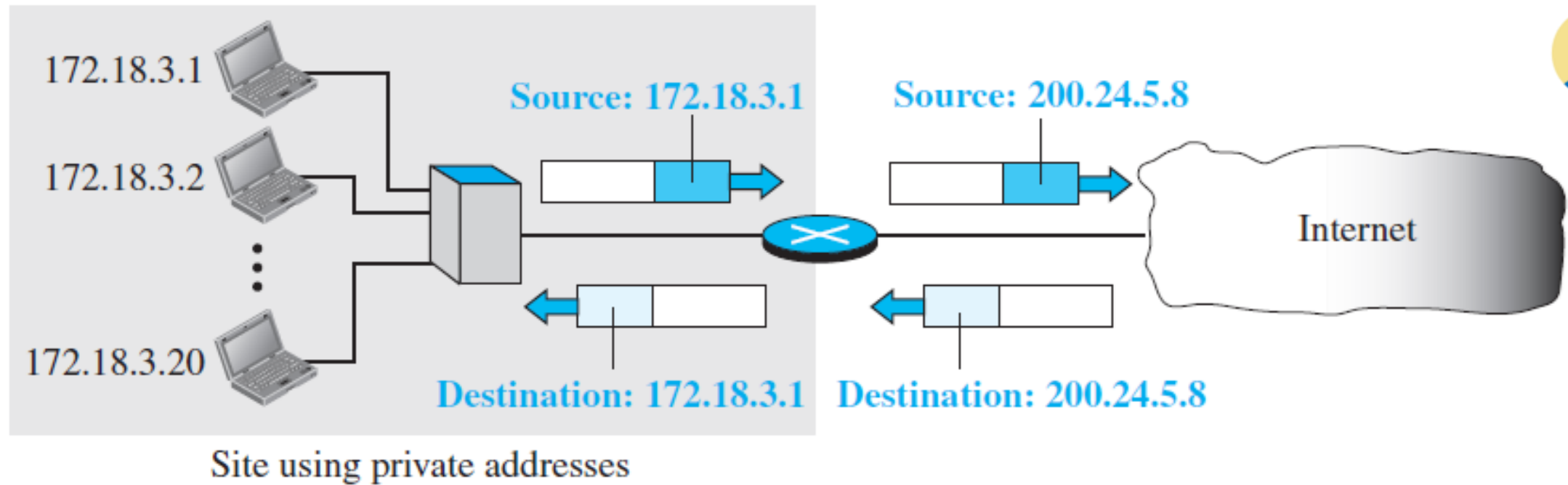


NETWORK ADDRESS TRANSLATION (NAT)

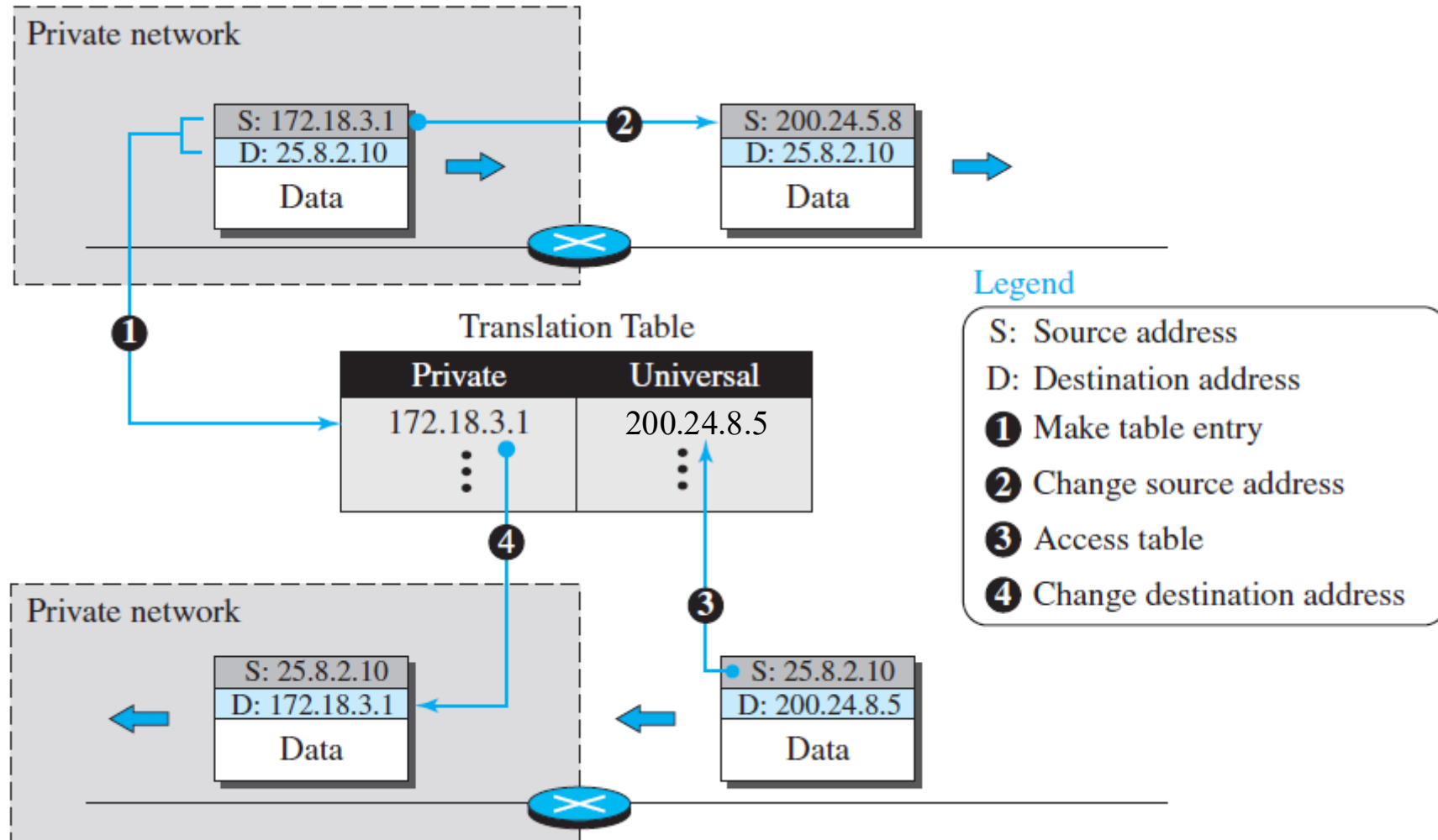


The technology allows a site to use a **set of private addresses** for internal communication and a **set of global Internet addresses** (at least one) for communication with the rest of the world.

ADDRESS TRANSLATION



TRANSLATION TABLE



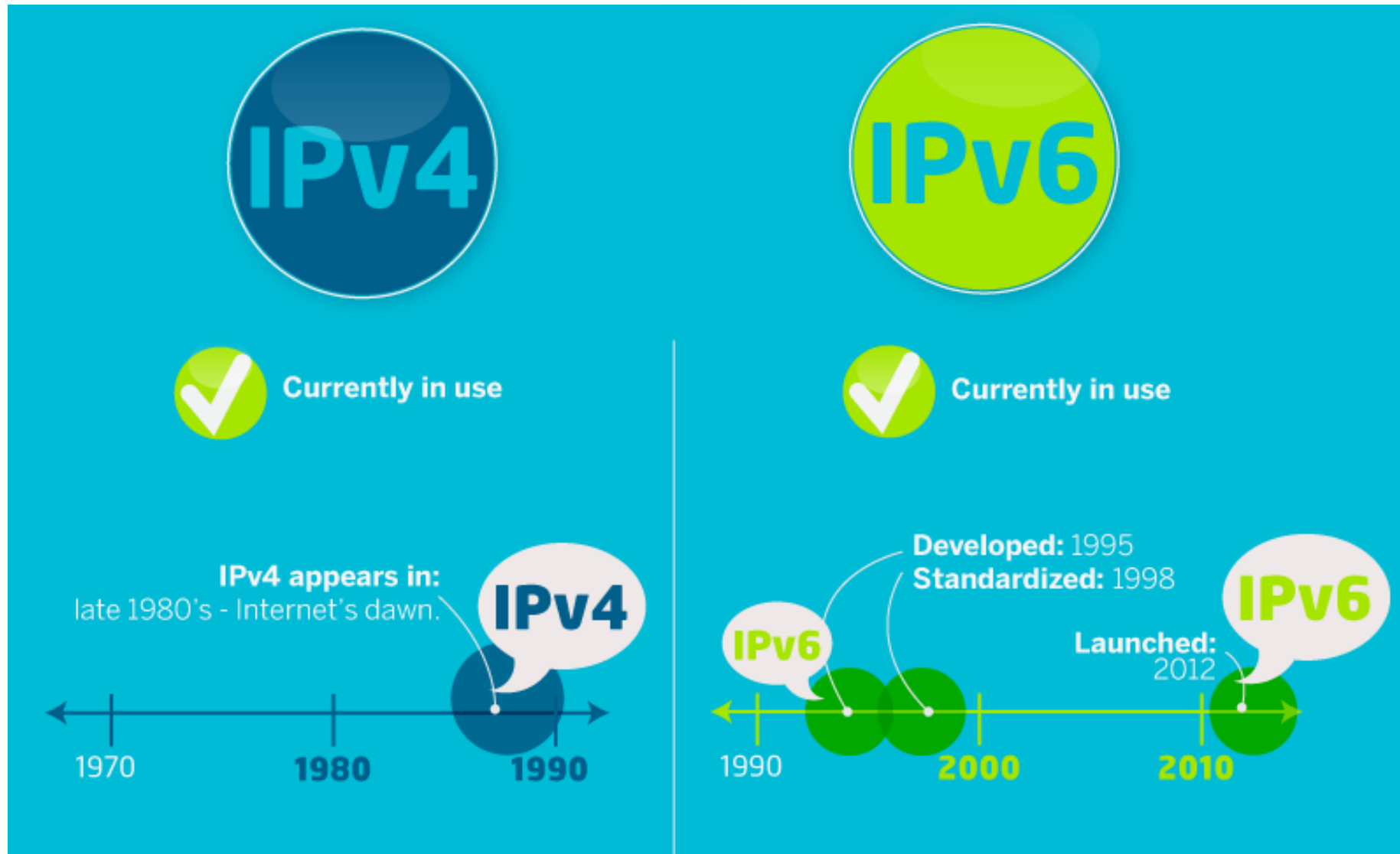
NAT

- Using a Pool of IP addresses:
 - The use of only one global address allows only one private-network host to access a given external host.
 - To remove this restriction, the NAT router can use a pool of global addresses.
- Using Both IP Addresses and Port Addresses:

<i>Private address</i>	<i>Private port</i>	<i>External address</i>	<i>External port</i>	<i>Transport protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
⋮	⋮	⋮	⋮	⋮



IPv6



IPV6

- Representation

Binary (128 bits)	111111011110110 ... 111111100000000
Colon Hexadecimal	FEF6:BA98:7654:3210:ADEF:BBFF:2922:FF00

- Abbreviation – Zero Compression

FDEC:0:0:0:0:BBFF:0:FFFF → **FDEC::BBFF:0:FFFF**

- Mixed Notation

- Colon hex notation for the leftmost six sections and four-byte dotted-decimal notation instead of the rightmost two sections

0:0:0:0:0:0:13.1.68.3

- CIDR Notation

FDEC::BBFF:0:FFFF/60



IPV6 ADDRESS SPACE

Enter **IPv6**. The number of possible IPv6 addresses is
340,282,366,920,938,463,463,374,607,431,768,211,456.

(340 UNDECILLION)

We can assign an IP address to every atom on
the surface of the Earth, and still have enough left over
for a few other planets.

IPv6

=

3.4×10^{38}
network
addresses



IPV6 THREE ADDRESS TYPES

- Unicast Address (One-to-One)
 - identifies one specific interface/device.
- Anycast Address (One-to-One of Many)
 - assigned to multiple devices, but the packet is delivered to only one of them, normally the nearest/best device according to the routing system.
- Multicast Address (Many-to-Many)
 - identifies a group of devices.

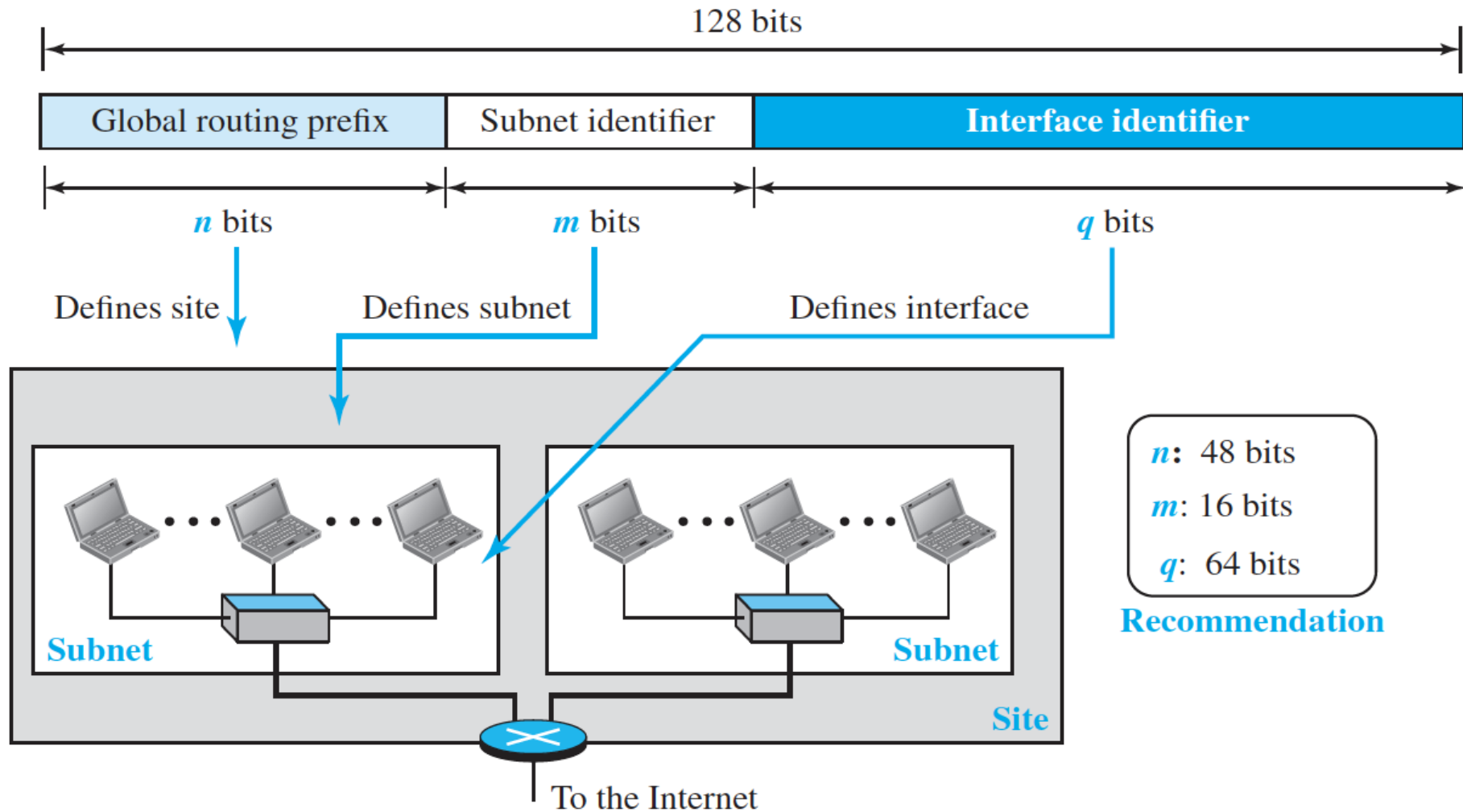


ADDRESS SPACE ALLOCATION

<i>Block prefix</i>	<i>CIDR</i>	<i>Block assignment</i>	<i>Fraction</i>
0000 0000	0000::/8	Special addresses	1/256
001	2000::/3	Global unicast	1/8
1111 110	FC00::/7	Unique local unicast	1/128
1111 1110 10	FE80::/10	Link local addresses	1/1024
1111 1111	FF00::/8	Multicast addresses	1/256



GLOBAL UNICAST ADDRESS

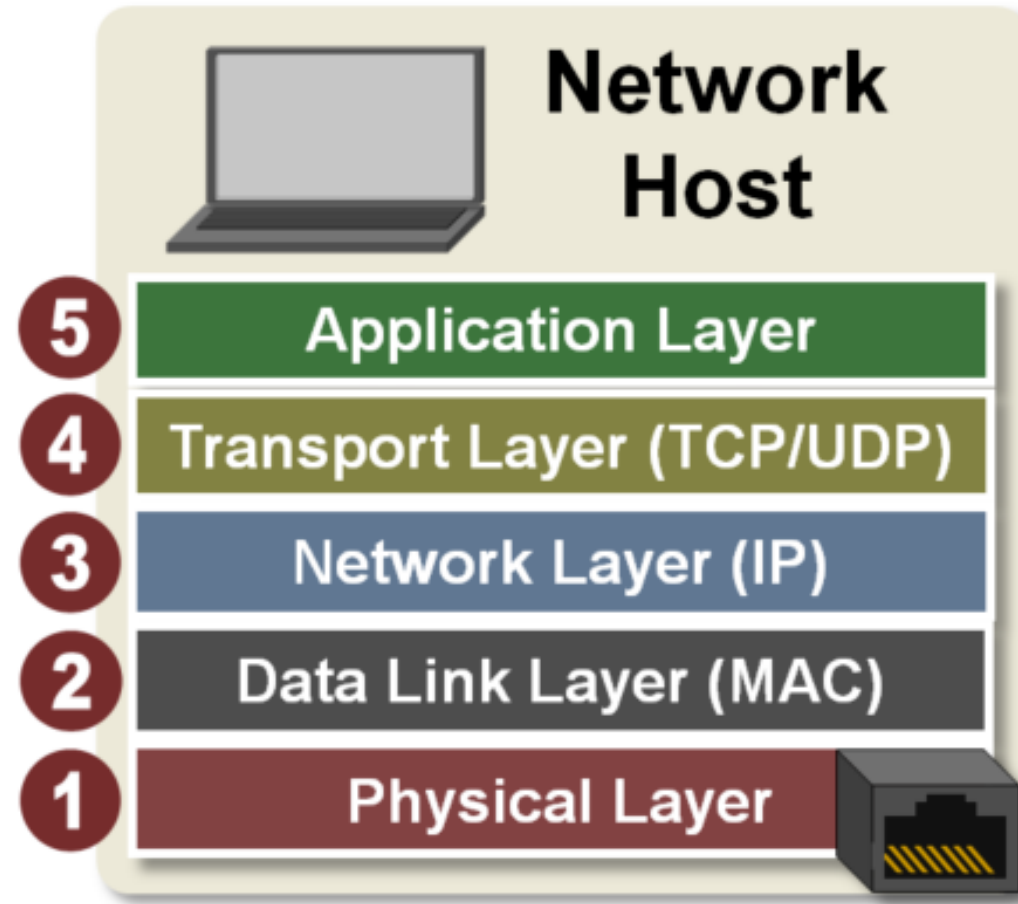




TRANSPORT LAYER

Unit 4

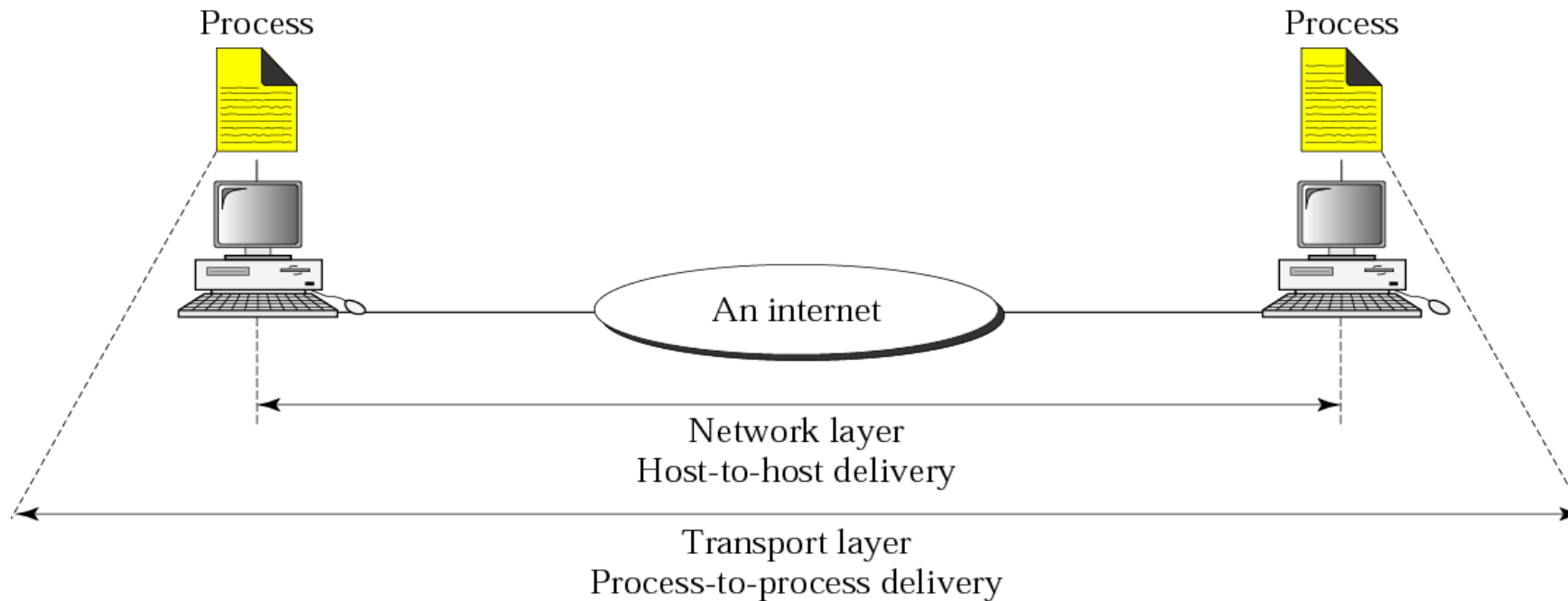
TRANSPORT LAYER





TRANSPORT LAYER SERVICES

PROCESS TO PROCESS COMMUNICATION



ADDRESSING: PORT NUMBERS

- The most common way to achieve process to process communication is through the **client-server paradigm**
 - Local Host – Client; Remote Host – Server
- OS today support both **multiuser and multiprogramming environments**
- The local host and the remote host are defined using **IP addresses**
- To define the processes, we need second identifiers, called **port numbers (in the range 0 to 65,535)**



ADDRESSING: PORT NUMBERS

- The client program defines itself with a port number, called the **ephemeral port number**.
- Recommended to be **greater than 1023** for some client/server programs to work properly.
- The server process must also define itself with a port number however, it cannot be chosen randomly.
- TCP/IP has decided to use universal port numbers for servers called **well-known port numbers**.

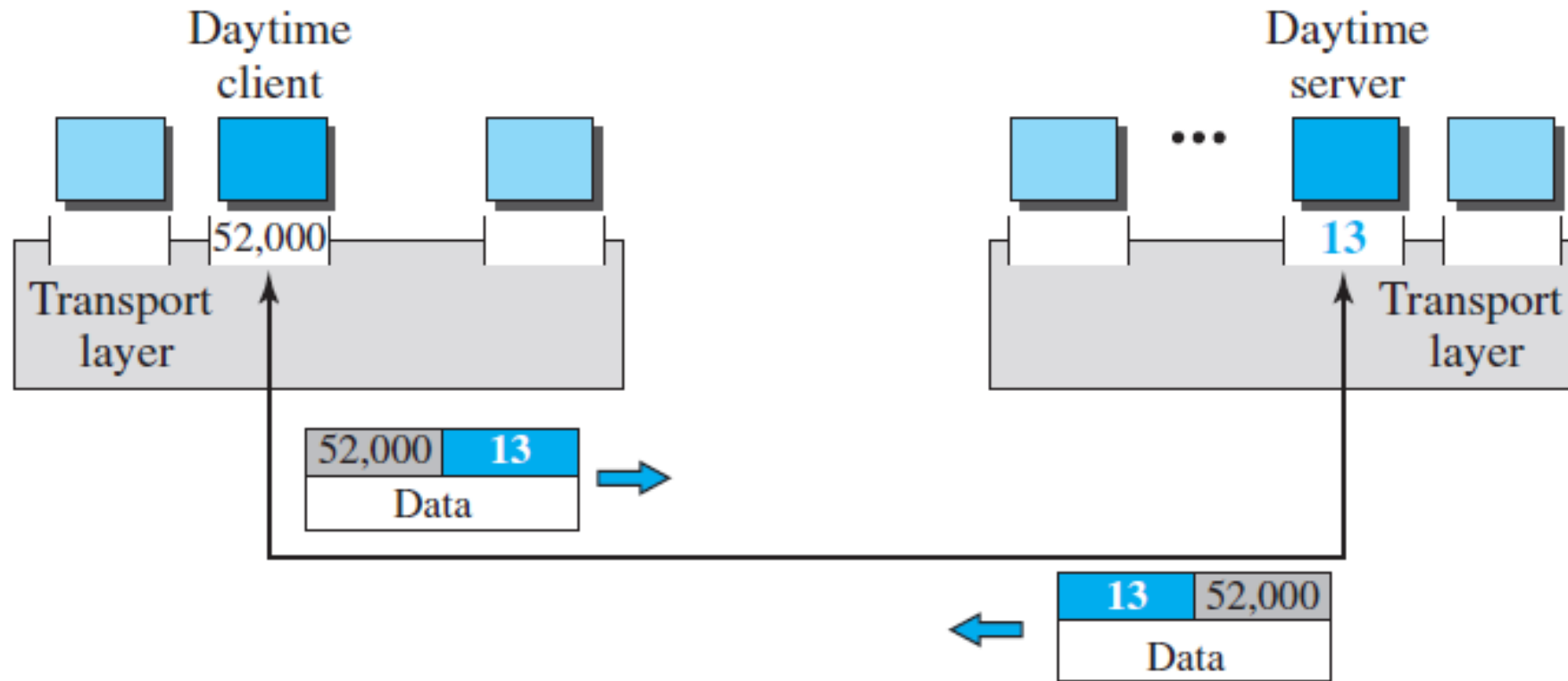


WELL KNOWN PORT NUMBERS

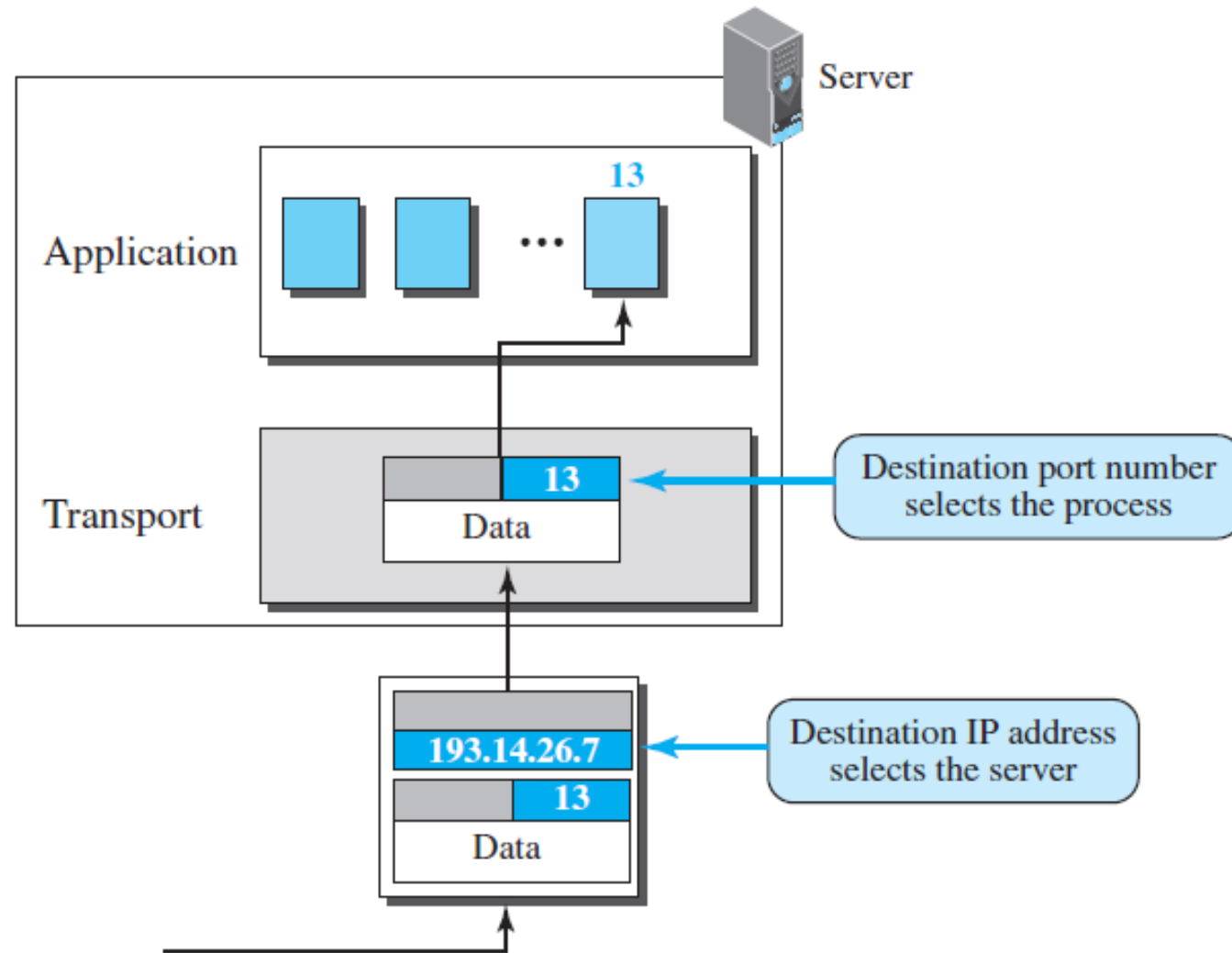


Port Number	Application Protocol	Description	Transport Protocol
21	FTP	File transfer	TCP
23	Telnet	Remote login	TCP
25	SMTP	E-mail	TCP
53	DNS	Domain Name System	UDP
79	Finger	Lookup information about a user	TCP
80	HTTP	World wide web	TCP
110	POP-3	Remote e-mail access	TCP
119	NNTP	Usenet news	TCP
161	SNMP	Simple Network Management Protocol	UDP

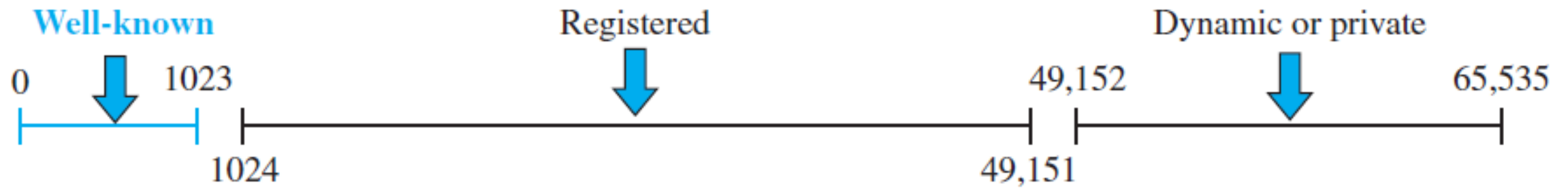
ADDRESSING: PORT NUMBERS



IP ADDRESS AND PORT NUMBER



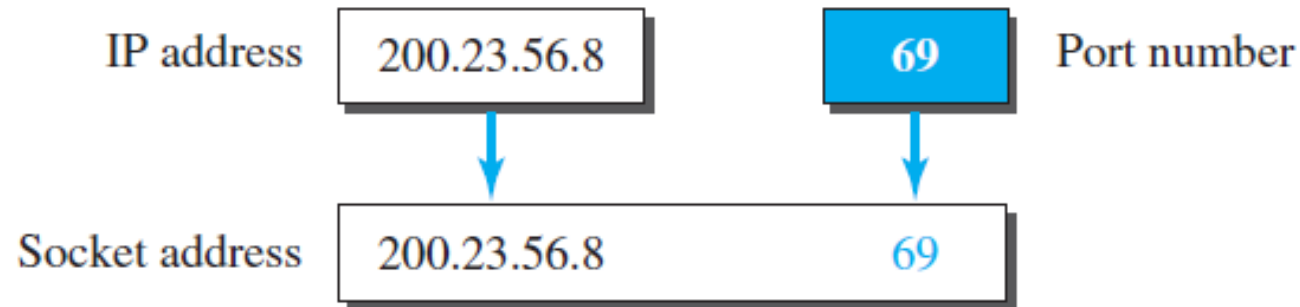
ICANN RANGES



ICANN: Internet Corporation for Assigned Names and Numbers

SOCKET ADDRESSES

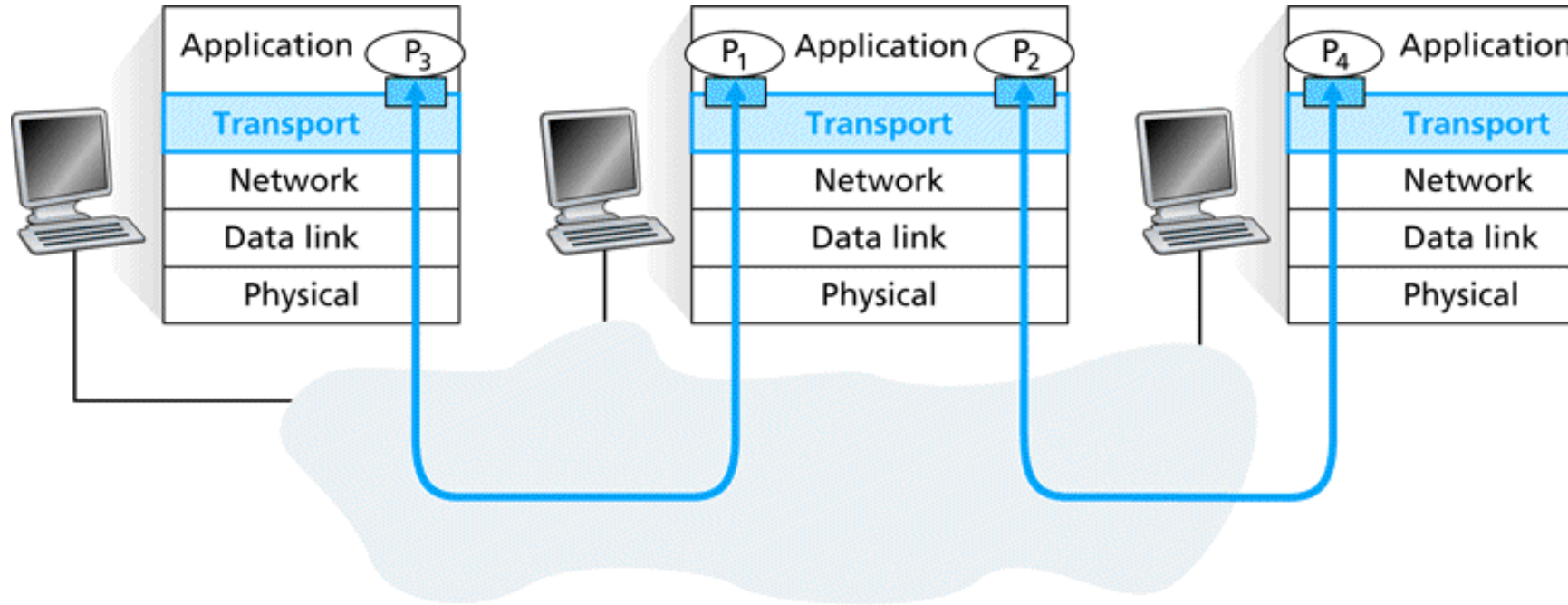
- Combination of an IP address and a port number



- A pair of socket addresses: the client socket address and the server socket address are required.
- These four pieces of information are part of the network-layer packet header and the transport-layer packet header.



SOCKET ADDRESSES



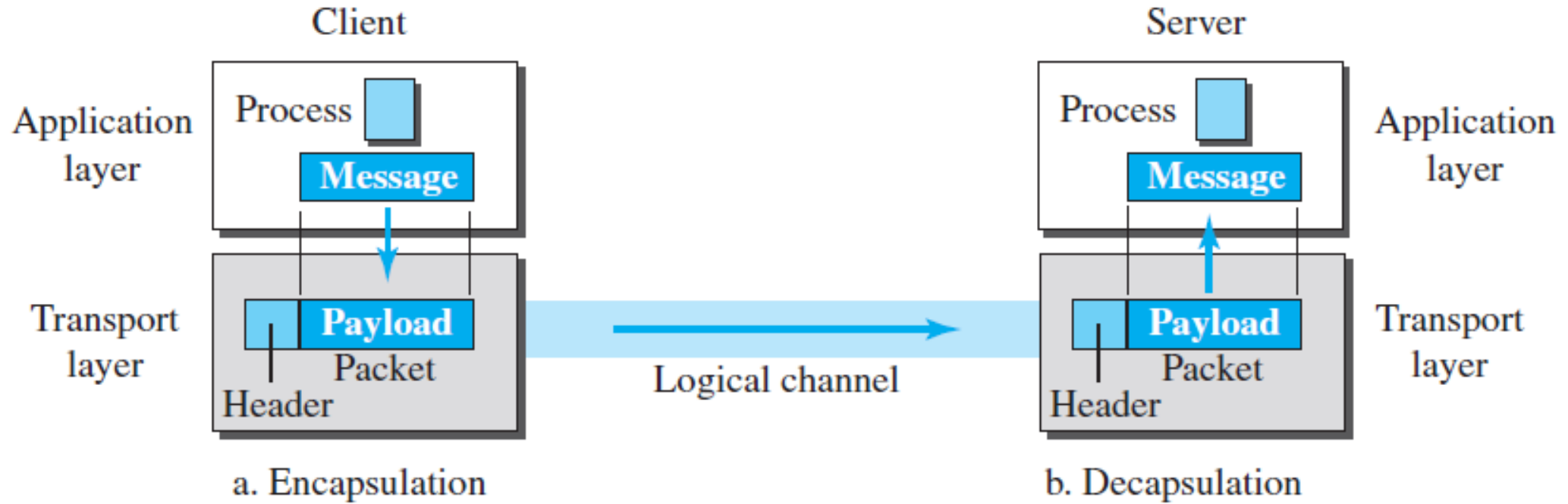
Key:

○ Process

■ Socket



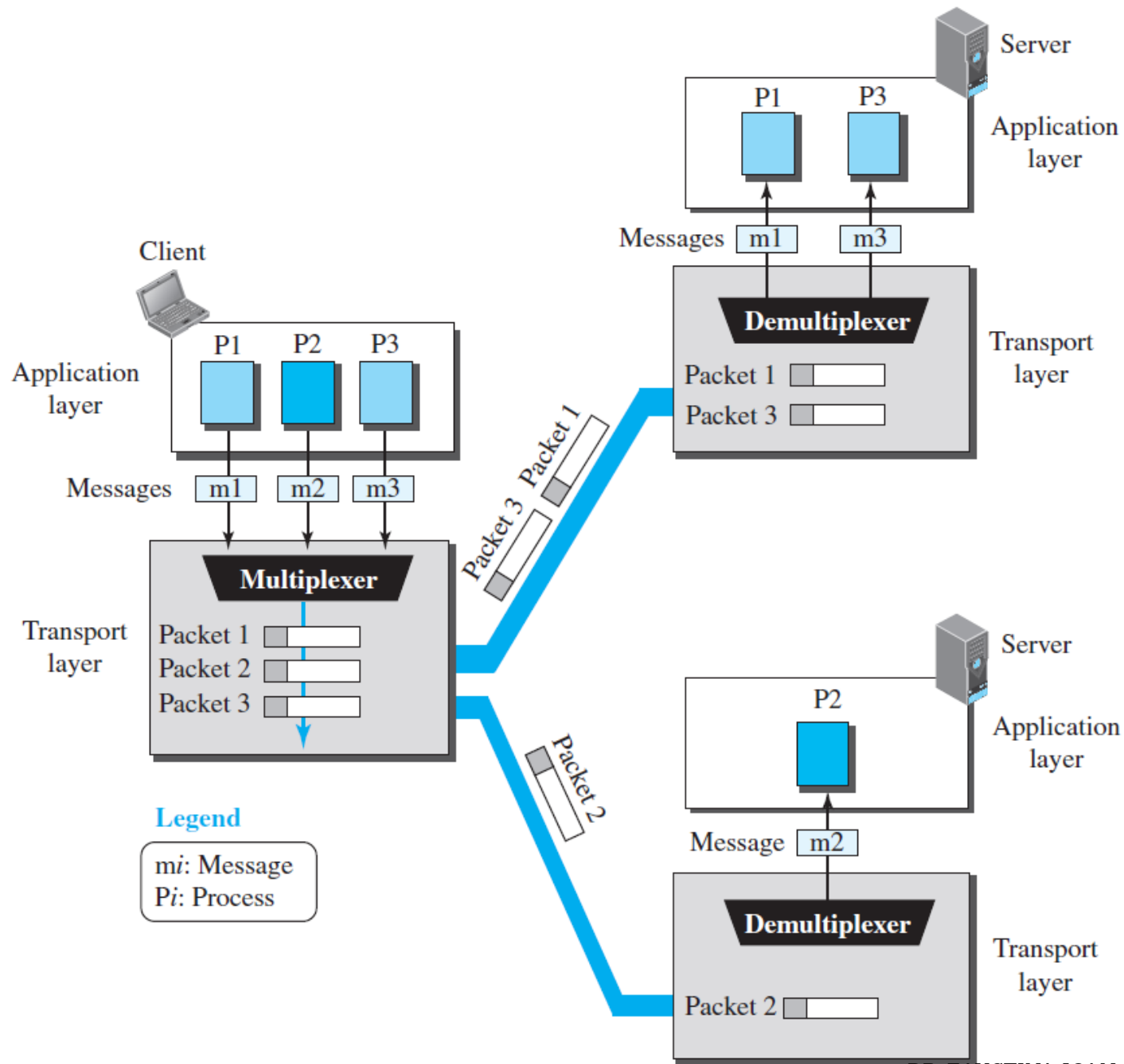
ENCAPSULATION AND DECAPSULATION



MULTIPLEXING AND DEMULTIPLEXING

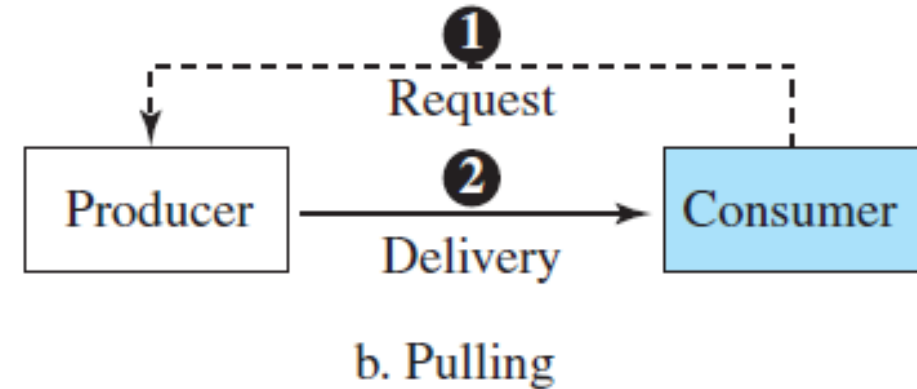
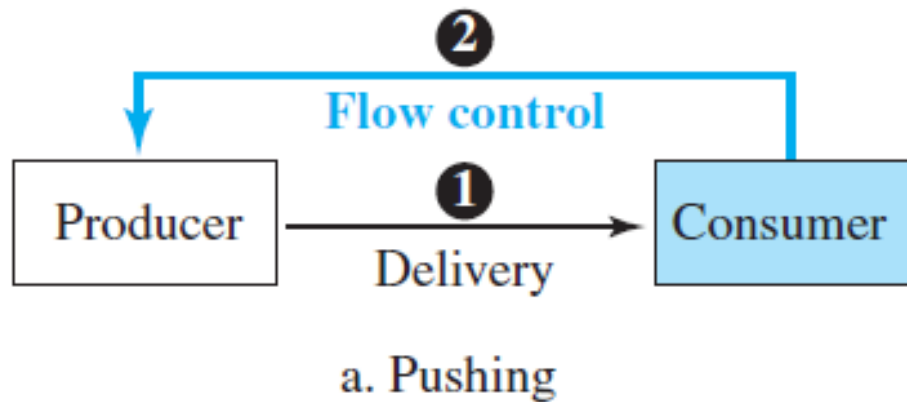
- Whenever an entity accepts items from more than one source – **multiplexing (many to one)**
- Whenever an entity delivers items to more than one source – **demultiplexing (one to many)**





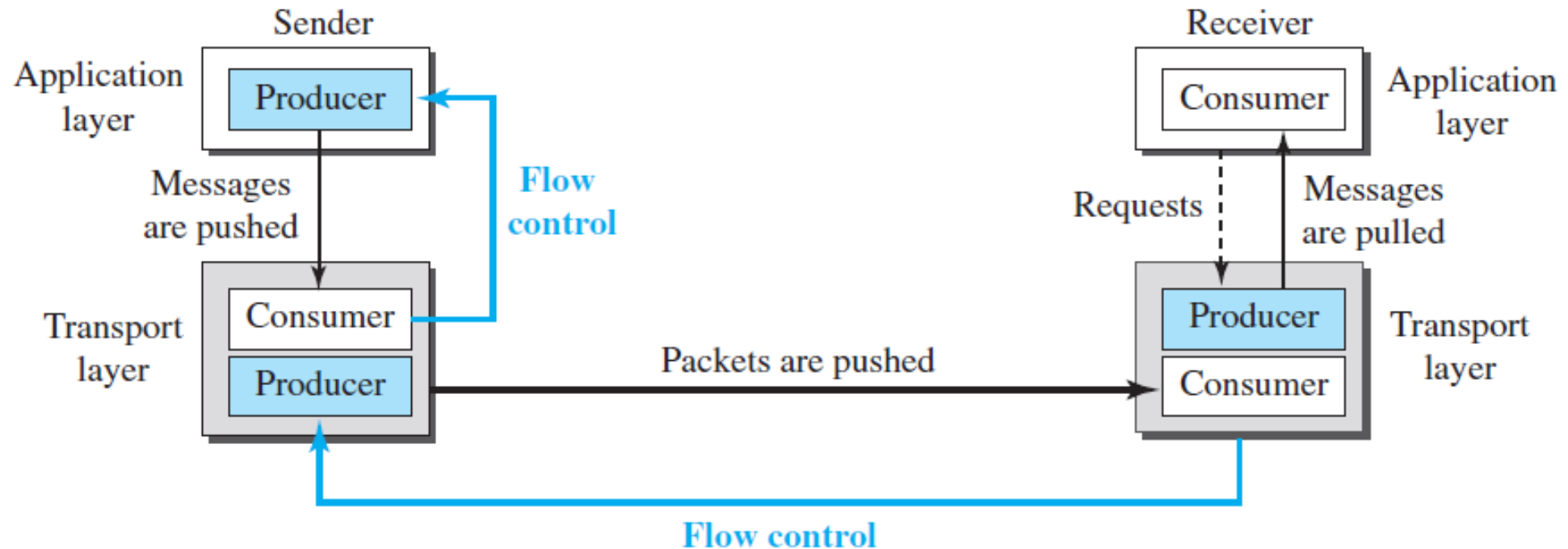
FLOW CONTROL

- Delivery of items from a producer to a consumer can occur in one of two ways: **pushing or pulling**.



FLOW CONTROL AT TRANSPORT LAYER

- Four entities for communication: sender process, sender transport layer, receiver transport layer, and receiver process.

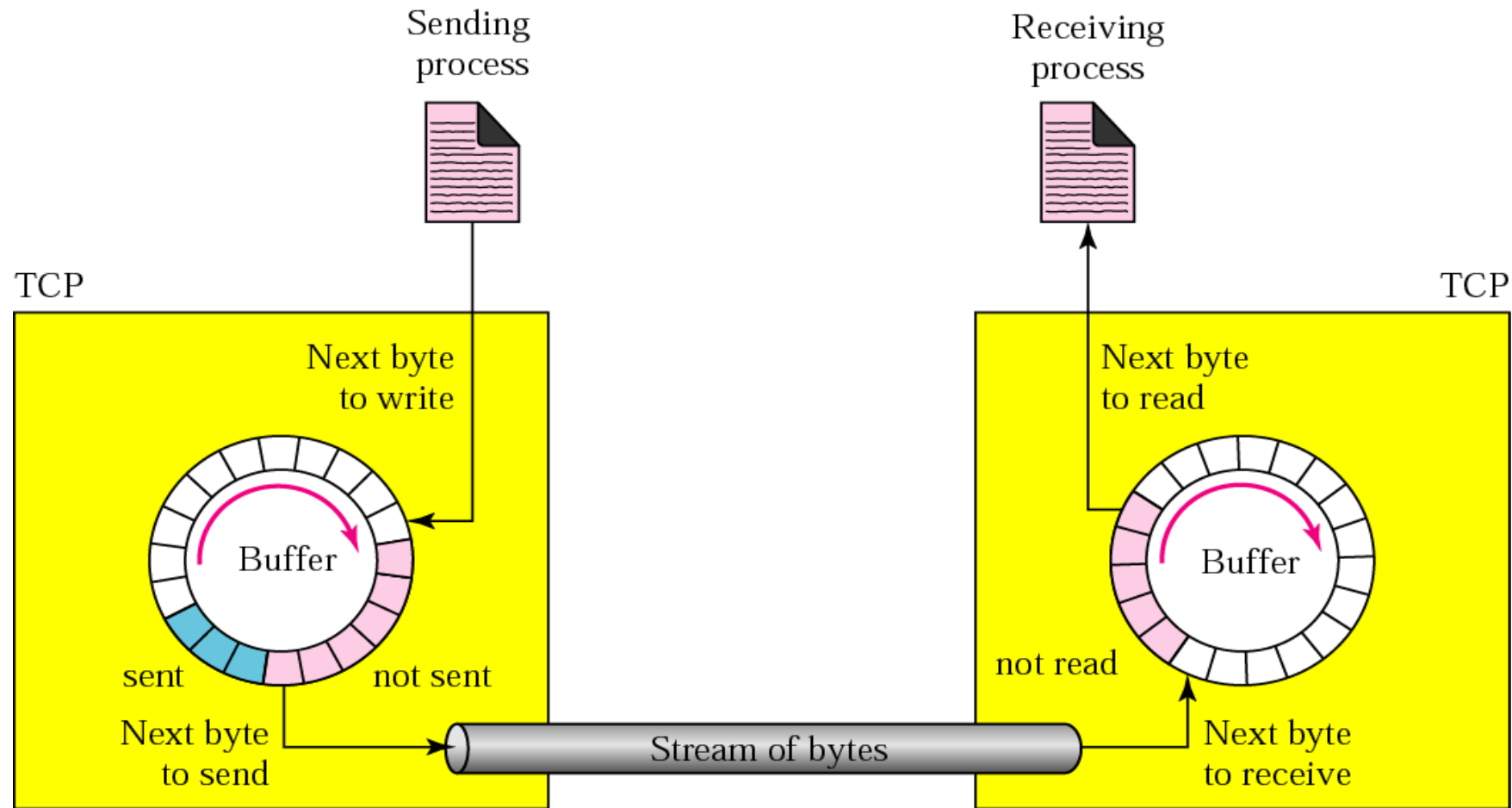


FLOW CONTROL AT TRANSPORT LAYER

- Although flow control can be implemented in several ways, one of the solutions is normally to use **two buffers**:
 - One at the sending transport layer and the other at the receiving transport layer.
- A buffer is a set of memory locations that can hold packets at the sender and receiver.



FLOW CONTROL AT TRANSPORT LAYER



ERROR CONTROL

- In the Internet, the underlying network layer (IP) is unreliable.
- The transport layer should be made reliable if the application requires reliability.
- Reliability can be achieved to add error control services to the transport layer.



ERROR CONTROL

- Error control at the transport layer is responsible for
 1. Detecting and discarding corrupted packets.
 2. Keeping track of lost and discarded packets and resending them.
 3. Recognizing duplicate packets and discarding them.
 4. Buffering out-of-order packets until the missing packets arrive.



ERROR CONTROL

- Error control requires that the sending transport layer to know
 - which packet is to be resent
 - which packet is a duplicate, or
 - which packet has arrived out of order
- Solution: Add a field to the transport-layer packet to hold the **sequence number** of the packet.



SEQUENCE NUMBERS

- Packets are numbered sequentially.
- However, because it needs to be included in each packet in the header, a limit should be set.
- If the header of the packet allows m bits for the sequence number, the sequence numbers range from 0 to $2^m - 1$.
- If $m=4$, only sequence numbers 0 to 15 are allowed

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, ..

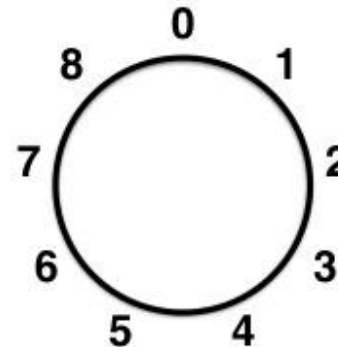
- In other words, the sequence numbers are modulo 2^m .



SEQUENCE NUMBERS

Modulus 9

$0 \bmod 9 = 0$	$9 \bmod 9 = 0$
$1 \bmod 9 = 1$	$10 \bmod 9 = 1$
$2 \bmod 9 = 2$	$11 \bmod 9 = 2$
$3 \bmod 9 = 3$	$12 \bmod 9 = 3$
$4 \bmod 9 = 4$	$13 \bmod 9 = 4$
$5 \bmod 9 = 5$	$14 \bmod 9 = 5$
$6 \bmod 9 = 6$	$15 \bmod 9 = 6$
$7 \bmod 9 = 7$	$16 \bmod 9 = 7$
$8 \bmod 9 = 8$	$17 \bmod 9 = 8$



ACKNOWLEDGMENT

- Both positive and negative signals can be used as error control, but we discuss only positive signals, which are more common at the transport layer.



ACKNOWLEDGEMENT - RECEIVER'S SIDE

- The receiver side can send an acknowledgment (ACK) for each of a collection of packets that have arrived safe and sound.
- The receiver can simply discard the corrupted and duplicate packets.

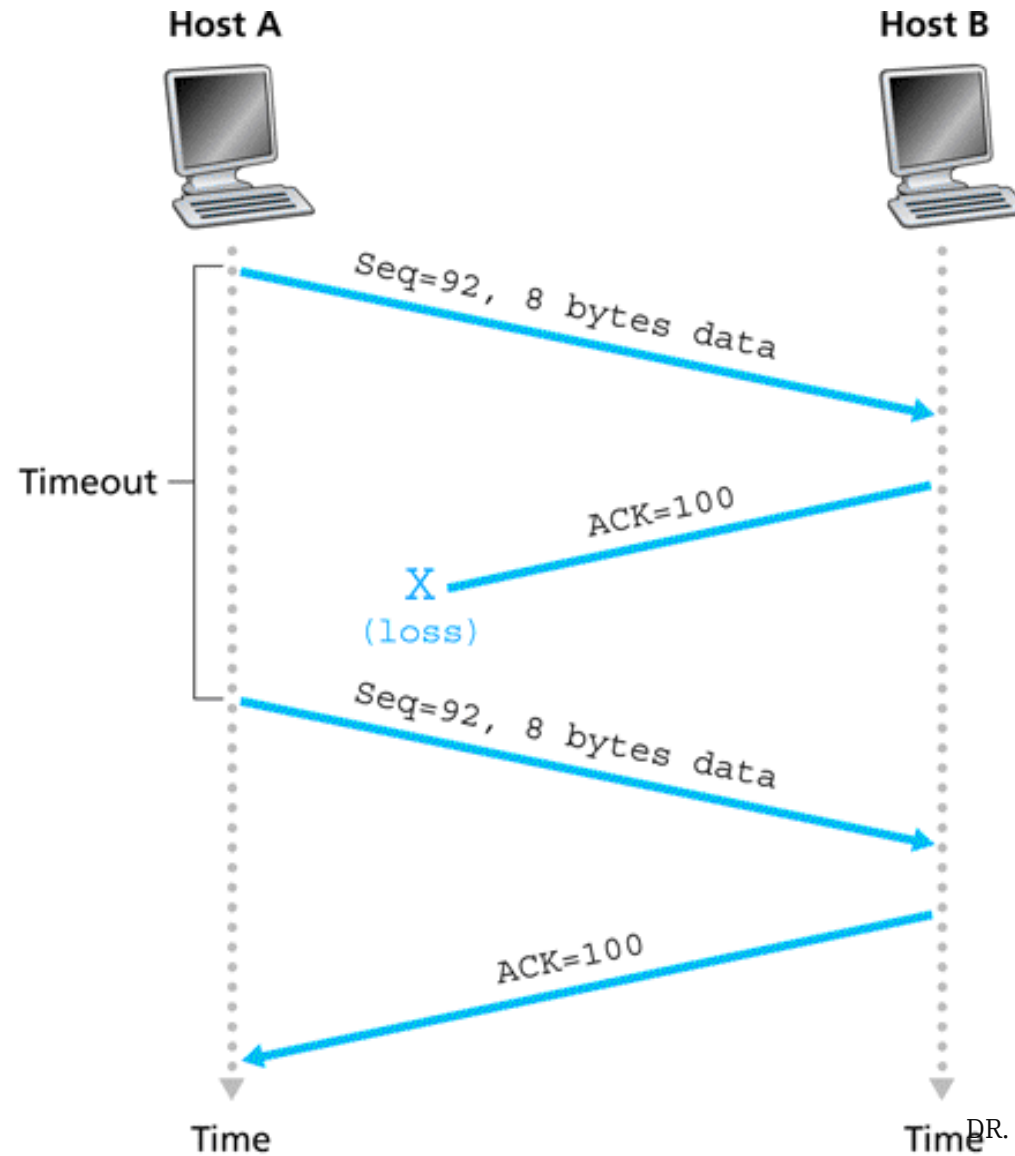


ACKNOWLEDGEMENT - SENDER'S SIDE

- The sender can detect lost packets if it uses a timer.
- When a packet is sent, the sender starts a timer.
- If an ACK does not arrive before the timer expires, the sender resends the packet.
- Out-of-order packets can be either discarded (to be treated as lost packets by the sender), or stored until the missing one arrives.



ACKNOWLEDGMENT

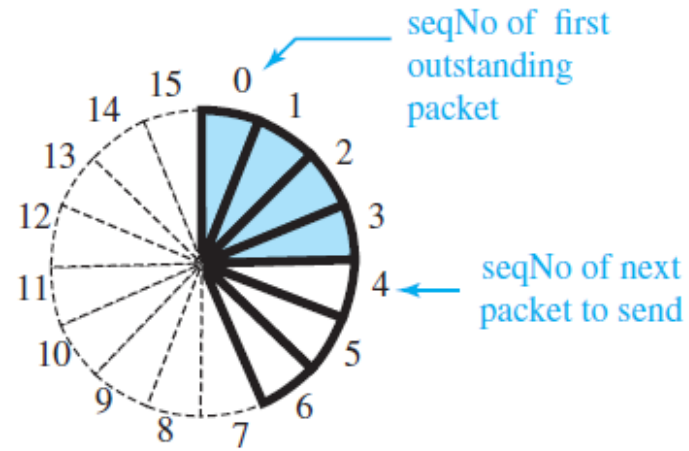


COMBINATION OF FLOW AND ERROR CONTROL

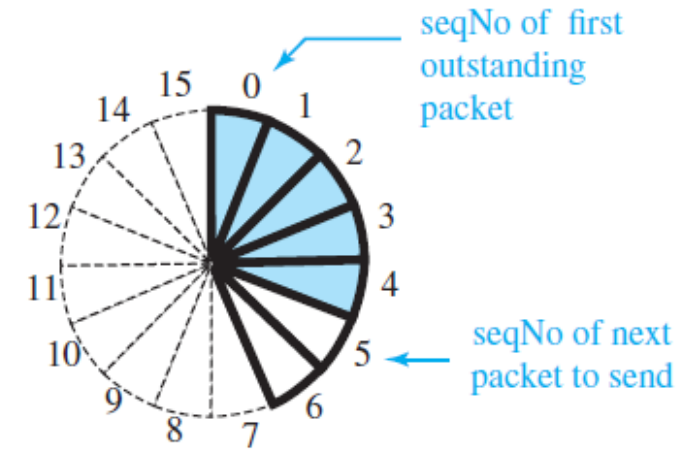
- Flow control requires the use of two buffers, one at the sender site and the other at the receiver site.
- Error control requires the use of sequence and acknowledgment numbers by both sides.
- Combination of both: use **two numbered buffers**, one at the sender, one at the receiver.



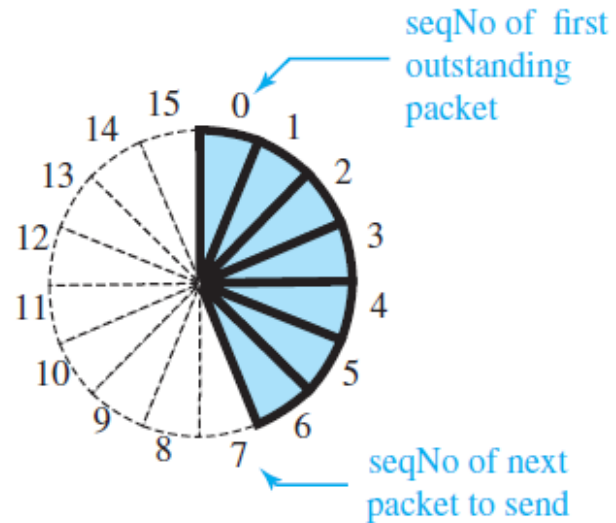
SLIDING WINDOW



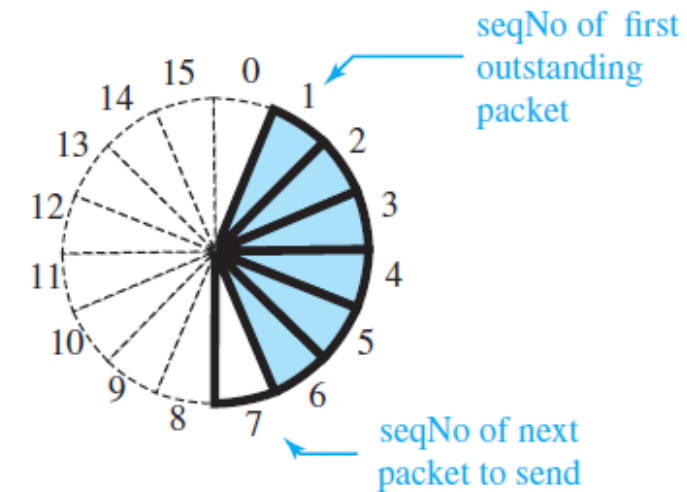
a. Four packets have been sent.



b. Five packets have been sent.



c. Seven packets have been sent;
window is full.



d. Packet 0 has been acknowledged;
window slides.



AT SENDER'S SIDE

- When a packet is to be sent, the number of the next free location, x , in the buffer is used as the sequence number of the packet.
- When the packet is sent, a copy is still there at location x , awaiting the acknowledgment from the other end.
- When the acknowledgment arrives, the packet is removed and the memory location becomes free.



AT RECEIVER'S SIDE

- When a packet with sequence number y arrives, it is stored at the memory location y until the application layer is ready to receive it.
- An acknowledgment can be sent to announce the arrival of packet y .



SLIDING WINDOW - LINEAR



a. Four packets have been sent.



b. Five packets have been sent.



c. Seven packets have been sent;
window is full.



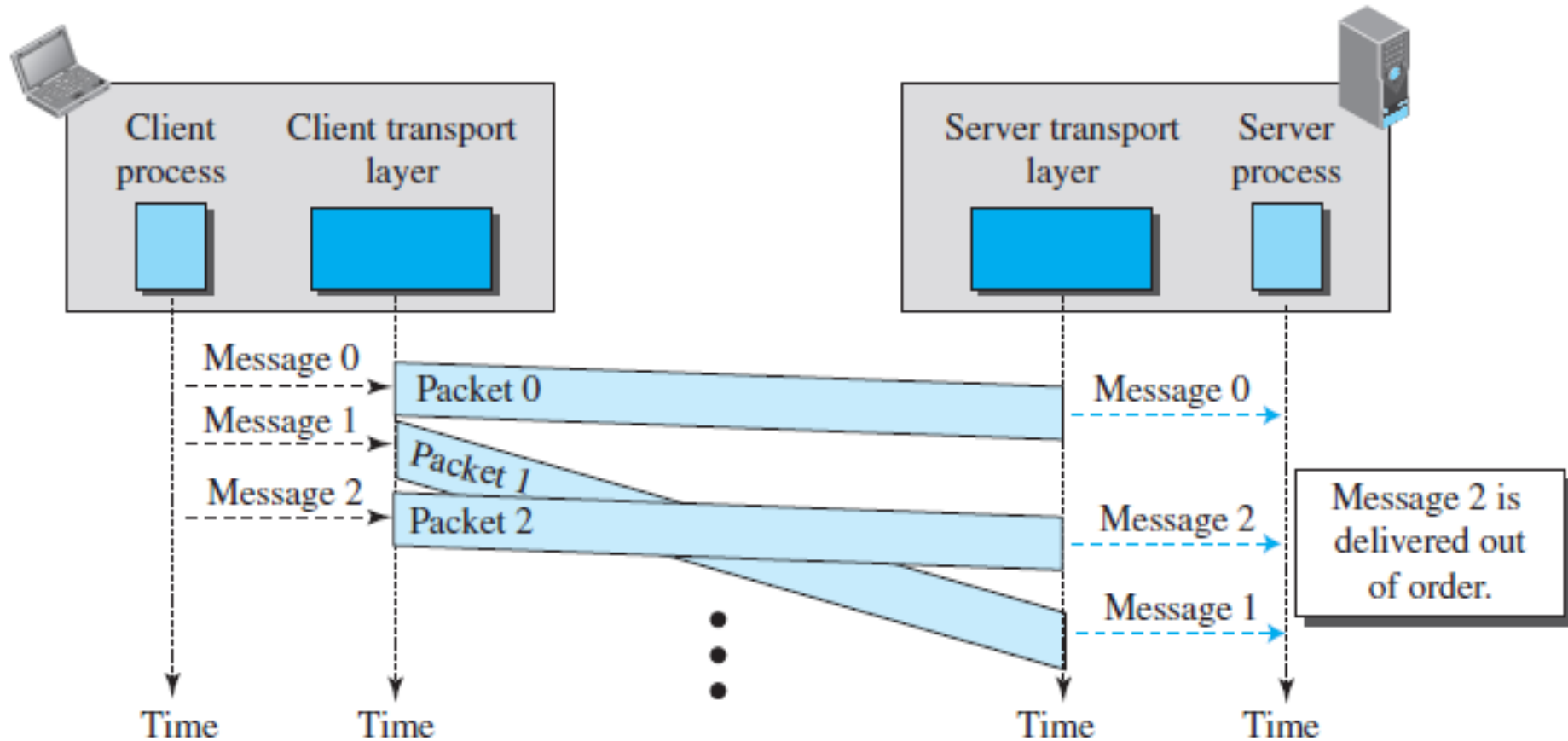
d. Packet 0 has been acknowledged;
window slides.

CONGESTION CONTROL

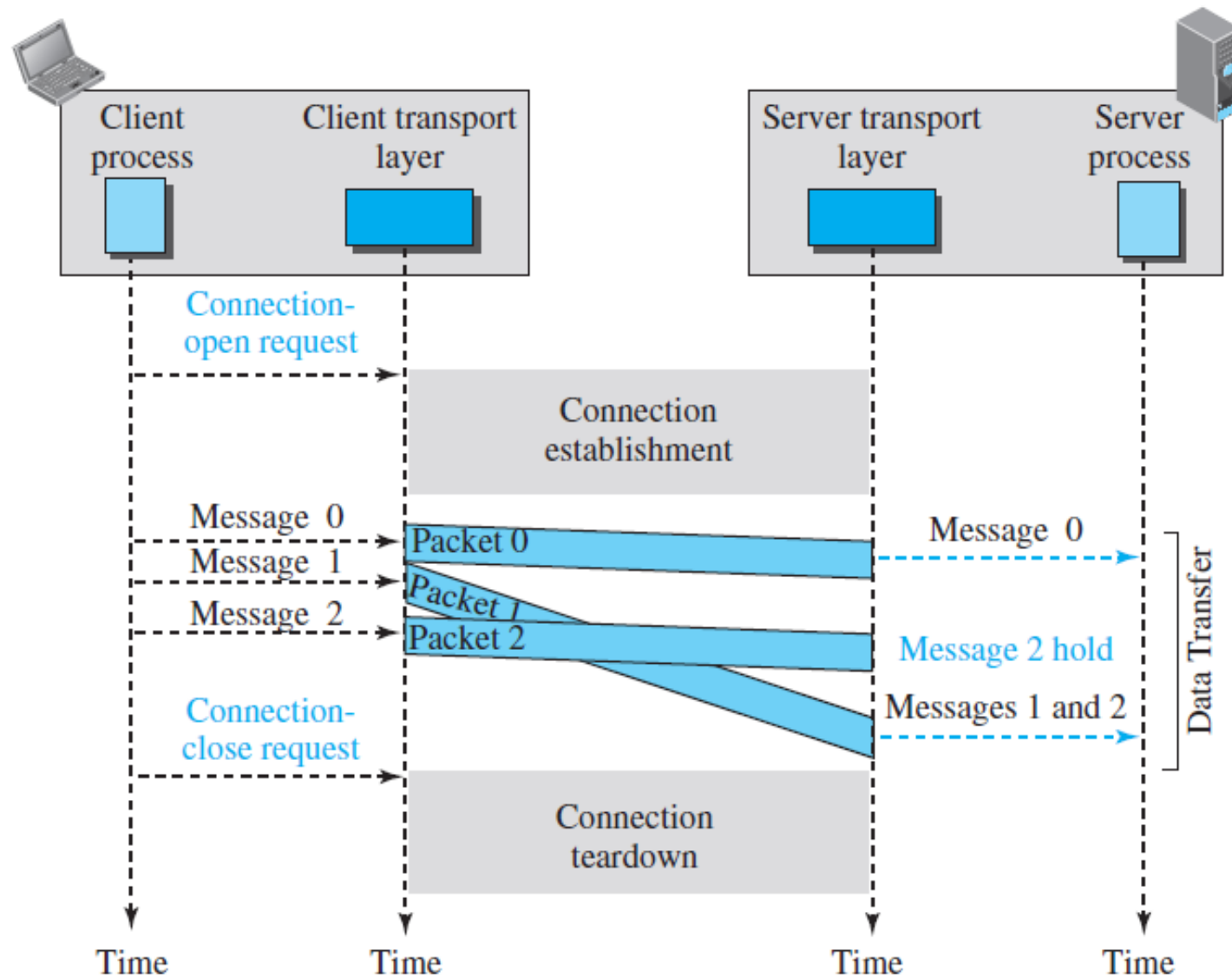
- Congestion in a network may occur if the **load** on the network is greater than the **capacity**.
- **Congestion control** refers to the mechanisms and techniques that control the congestion and **keep the load below the capacity**.
- Congestion at the transport layer is actually the result of congestion at the network layer



CONNECTIONLESS SERVICE



CONNECTION-ORIENTED SERVICE



Note:
The colored
arrow shows the
starting state.

